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Towards GDPR-Compliant Pseudonymization of Danish Psychotherapy Transcripts Using Named Entity Recognition

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Abstract

Psychotherapy transcripts contain sensitive personal information that must be anonymized before clinical analysis and research use. This work investigates whether Named Entity Recognition (NER) together with rule-based pseudonymization can effectively anonymize Danish psychotherapy transcripts while preserving privacy under GDPR constraints. A domain-relevant annotation schema inspired by GDPR-related privacy considerations is introduced together with an anonymization pipeline, consisting of two stages: entity detection and pseudonymization. Several approaches for entity detection were evaluated, such as Danish SpaCy NER models, a commercial PII detection system, few-shot prompting, and fine-tuned XLM-RoBERTa transformer models, including a LoRA-adapted variant. Results show that the fine-tuned XLM-RoBERTa models achieved the best overall performance, followed by the commercial PII model, whereas SpaCy models and few-shot prompting approaches performed poorly on psychotherapy transcripts, particularly for domain-specific entities. All approaches struggled with under-represented categories, such as demographic and health-related entities. Overall, the findings suggest that an NER-based anonymization pipeline is a promising framework for Danish psychotherapy transcripts, but current models remain insufficiently reliable for fully automated anonymization of sensitive clinical data.

Keywords: psychotherapy data, Named Entity Recognition (NER), pseudonymization, General Data Protection Regulation (GDPR), Danish clinical NLP

1 Introduction

Psychotherapy session transcripts are highly confidential texts. To make such valuable corpora usable for research, they must first be properly and reliably anonymized. This can be done in three levels: anonymization, de-identification, and pseudonymization. Anonymization is defined as an irreversible process in which data are transformed so that no information can be linked back to a specific individual (Meystre et al., 2010). Often, de-identification is used interchangeably with anonymization. However, de-identification focuses solely on removing direct identifiers, while anonymization must also address indirect identifiers (Baroud et al., 2025). In our project, we focus on pseudonymization. Pseudonymization aims to protect identities by replacing identifiers while keeping the remaining data available for analysis. This is typically achieved by storing identifying information separately from the data used for research, allowing the data to remain useful while reducing the risk of linking it to specific individuals (Attieh et al., 2025). This project is a part of a large project called ALF, which focuses on developing an LLM-based program that can accurately transcribe, anonymize, and annotate Danish psychotherapy sessions. In this specific project, we focus on the anonymization (pseudonymization) part.

While the psychotherapy session transcripts contain personally identifiable information, such as direct identifiers (e.g., names, possi-

bly addresses, and CPR numbers) (Pilán et al., 2022), they also include more subtle, highly domain-specific identifiers that, in combination with other identifiers, could identify an individual. These are called quasi-identifiers. They are, as mentioned before, attributes that do not directly identify an individual but may enable re-identification when combined with other information (Pilán et al., 2022), such as workplaces, family relations, hospitals or clinics, and others. Furthermore, specialized methods may be required for anonymization of sensitive attributes such as health conditions and other demographic attributes (Khan et al., 2025). As a result, an anonymization in this setting is significantly more complex than in general domain-specific text. It requires a model that can capture both explicit mentions and context-dependent identifiers to prevent privacy leakage and potential legal consequences.

In this project, we apply Named Entity Recognition (NER) to the ethically sensitive domain of Danish psychotherapy session transcripts. NER is a core technique within Natural Language Processing (NLP) that enables automatic extraction of information from unstructured text, such as these transcripts. It allows models to detect and classify entities such as persons, locations, organizations, and other sensitive information directly from the raw text (Pamarthi, 2024). A further challenge is the linguistic setting. The transcripts are in Danish, where pretrained NER resources are more limited compared to English. This makes domain adaptation and careful evaluation particularly important.

To address these challenges, we compare four approaches that represent different levels of domain adaptation and data requirements:

- (i) an off-the-shelf Danish NER model,
- (ii) a commercial PII detection system,
- (iii) few-shot prompting approaches, and
- (iv) fine-tuned transformer-based models

trained on annotated data.

The purpose of this comparison is to examine how different levels of adaptation and specialization influence anonymization performance in highly sensitive, domain-specific settings. We analyze the different types of errors these models make and compare them to one another.

1.1 Background and Motivation

Mental health disorders are a common issue for many people in today’s world (World Health Organization, 2022), yet many of the mechanisms behind successful psychotherapy treatments remain poorly understood. Recent progress in artificial intelligence and large language models has opened new possibilities for analyzing psychotherapy sessions and identifying patterns linked to treatment prescription outcomes. However, this type of research requires access to large collections of psychotherapy data, which raises significant legal and ethical concerns given that clinical transcripts are of a highly sensitive nature (European Parliament and Council of the European Union, 2016a).

The information within the psychotherapy data may enable re-identification of patients (Pilán et al., 2022), and for this reason, a reliable anonymization process represents the main bottleneck in developing systems that can be fed with clinical psychotherapy data.

Our work was conducted within the broader ALF research project, which focuses on AI-assisted Danish psychotherapy research. The scope of the ALF project is to create a unified system capable of automatically transcribing, anonymizing, and annotating Danish psychotherapy sessions using large language models and automatic speech recognition systems. The platform is meant to support secure handling of sensitive data, and as a result of this, also allow for future AI-based mental health research.

Within the ALF pipeline, anonymization plays a crucial role, as insufficient anonymization of psychotherapy transcripts would make the entire process legally and ethically problematic. Therefore, the motivation behind our work lies in the support of the ALF project itself and also in contributing to the development of a scalable and reusable anonymization solution that can be applied to future clinical research projects.

1.2 Problem Statement and Research Question

Psychotherapy transcripts contain extremely sensitive personal information (e.g. names, locations, contact details) and are protected by the General Data Protection Regulation (GDPR), which severely limits their use for research or AI development. Therefore, anonymization is necessary for building clinically useful language models. While anonymization tools already exist, they are often trained on legal, general-purpose text, or structured medical records. These approaches do not translate well to psychotherapy transcripts, where sensitive information is often expressed casually, indirectly, or in the flow of conversation. This project addresses that gap by developing and evaluating an anonymization model specifically adapted to Danish psychotherapy transcripts.

1.2.1 Problem statement

Current anonymization models for Danish text are not validated on psychotherapy data and may fail to detect personally identifiable information (PII) in clinical dialogue, posing legal and ethical risks.

1.2.2 Research question

Can a pipeline combining named entity recognition (NER) and rule-based pseudonymization effectively anonymize Danish psychotherapy transcripts for

LLM training?

2 Related Work

Many recent papers dealt with the challenge of detecting and anonymizing sensitive information in text. Pilán et al. (Pilán et al., 2022) introduced the Text Anonymization Benchmark (TAB), which is a corpus and evaluation framework that supports systematic comparison of automated anonymization methods through manually annotated documents, standardized metrics and annotation guidelines. Continuing with the general anonymization problem, Vakili et al. (Vakili et al., 2023) focused on the clinical domain, and investigated whether pseudonymization (replacing personally identifiable information (PII) by NER) can preserve utility in BERT-based clinical models. Their results on Swedish electronic health records suggest that pseudonymization can reduce privacy risks while maintaining model performance, highlighting the importance of balancing privacy protection with data utility. More work emphasized that detecting direct identifiers alone is insufficient for anonymization. Baroud et al. (Baroud et al., 2025) write about a structured approach for identifying indirect personal identifiers (IPIs) in medical discharge summaries. They show that detecting these indirect identifiers is way more challenging than detecting direct PHI, yet they can significantly contribute to re-identification risk when combined with external knowledge.

Together, these papers form the foundation for our project. TAB provides a baseline for designing our tag scheme, the clinical pseudonymization literature motivates our use of NER-based detection of PIIs in psychotherapy transcripts, and the work on indirect identifiers supports our decision to consider a broader set of sensitive entities, including contextual information.

2.1 The Text Anonymization Benchmark (TAB)

The Text Anonymization Benchmark (TAB) by Pilán et al. (Pilán et al., 2022) provides a corpus and an evaluation framework for text anonymization. The benchmark was designed to support the development and systematic evaluation of automated methods for detecting and anonymizing sensitive information in text data. TAB includes manually annotated documents containing various types of personal identifiers and sensitive information, enabling researchers to evaluate how effectively different approaches can detect and anonymize this information. In addition to annotation guidelines, the framework proposes evaluation procedures and metrics for comparing anonymization systems in a standardized way. They also highlight the challenges associated with protecting personal information in text while preserving the usefulness of data for analysis.

This paper is closely related to our project and served as a great baseline for making our own tag scheme, since it addresses the problem of identifying sensitive information in text data using NLP techniques. Both TAB and our work rely on the detection of personal identifiers and other sensitive entities that might reveal information about individuals. We could also say that our work complements this existing benchmark by exploring the performance of NER models in a specialized and sensitive application domain.

2.2 End-to-end pseudonymization of fine-tuned clinical BERT models

This paper by Vakili et al. (Vakili et al., 2023) investigates the impact of pseudonymization on the performance of clinical language models. Their work explores whether sensitive information in electronic health records (EHRs) can be replaced with surrogate values while preserving the usefulness of the data for NLP downstream tasks. The authors

apply pseudonymization using NER to detect personally identifiable information (PIIs) such as names, locations, and dates, and replace these entities with realistic alternatives of the same type, e.g. some of the most common names/surnames. They evaluate this approach on several clinical NLP tasks using the Swedish clinical datasets and BERT-based models, comparing models trained on original and pseudonymized data. They show that pseudonymization can reduce privacy risks while maintaining model performance.

These findings are particularly relevant to our work, as our project explores the use of NER-based techniques to identify sensitive entities in psychotherapy transcriptions. Our approach similarly relies on automatic detection of PIIs using NER. Furthermore, they highlight the challenge of balancing privacy protection and data utility, which is also important to our project.

2.3 Beyond De-Identification: A Structured Approach for Defining and Detecting Indirect Identifiers in Medical Texts

This paper by Baroud et al. (Baroud et al., 2025) emphasizes extending de-identification approaches by focusing on the identification of indirect personal identifiers (IPIs) in medical texts, in particular discharge summaries. Unlike standard PHI (protected health information) detection approaches, which primarily target direct identifiers such as names and addresses, this paper introduces identifying multiple categories of indirect identifiers, such as socio-economic status, lifestyle, family relations, and time information. As mentioned, the authors annotated a subset of discharge summaries using a predefined annotation schema defined in the paper. They show that identifying indirect identifiers is a lot more challenging than detecting direct PHIs, with a lower performance across models. They highlight that

indirect identifiers can significantly contribute to the re-identification risk when combined with external knowledge, hence emphasizing the need for more comprehensive anonymization strategies beyond traditional approaches.

This research project is very relevant to our work, as our project also goes beyond the simple detection of direct identifiers by considering a broader range of sensitive identifiers in psychotherapy transcripts. We also recognize that indirect or contextual information, such as health conditions, relationships, or personal circumstances, can contribute to re-identification risk. They also evaluated both BERT models and LLMs for the task of detecting direct identifiers, similarly to our project. This supports our motivation to evaluate and compare different approaches in a domain where indirect identifiers are frequent and important to ensure proper anonymization.

3 Data

3.1 Data Source and Collection

The data used in the project was collected by the VIRTU research team from the Mental Health Services in the Capital Region of Denmark (Region Hovedstadens Psykiatri, 2026). The data consists of psychotherapy sessions that were originally recorded in audio format as part of clinical treatment and research activities. The audio recordings were automatically transcribed as part of the ALF project using a fine-tuned Danish XLS-R Automatic Speech Recognition (ASR) system adapted for psychotherapy language (Hansen, 2022). The resulting transcripts were stored as .rttm (Rich Transcription Time Marked) files containing both spoken content and metadata such as timestamps and speaker identifiers.

In total, we work with 12 psychotherapy sessions, each with approximately 45 minutes (2700 seconds) of recorded content, as illustrated in Figure 1. The transcripts reflect conversational speech, where utterances vary from

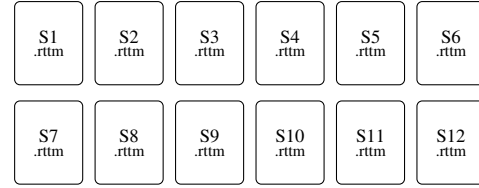


Figure 1: Overview of the 12 psychotherapy transcript sessions stored in .rttm format. Each session contains approximately 45 minutes of transcribed therapist-patient dialogue.

short responses (e.g., "Ja", "Okay") to full sentences. The dialogue between the patient and the therapist is interactive, meaning that the speakers can interrupt, pause, or reformulate their statements. Some sessions also include discussions related to Virtual Reality-based (VR) treatment, which can involve references to VR-headset devices, how to set them up etc., alongside the clinical dialogue.

3.2 Data Structure

Our data consists of psychotherapy session transcriptions derived from clinical recordings. Each transcript is structured as a sequence of utterances with associated metadata. To be specific, each entry contains a start and stop timestamp (in seconds), a binary speaker identifier (00 for therapist, 01 for patient), and the spoken text. An example of this structure is shown in Table 1.

Start	Stop	Speaker	Text
0.00s	2.80s	SPEAKER_00	Godmorgen, jeg hedder Anne Sørensen, og jeg er psykolog her på klinikken.
2.80s	6.10s	SPEAKER_01	Tak. Jeg hedder Thomas Nielsen.
6.10s	10.40s	SPEAKER_00	Vi har en tid klokken 09:30 i dag den 12. marts 2024.
10.40s	15.20s	SPEAKER_01	Ja, jeg bor i Aarhus på Nørrebrogade 14.
15.20s	18.60s	SPEAKER_00	Er det stadig postnummer 8000?
18.60s	23.50s	SPEAKER_01	Ja, det er det.
23.50s	28.20s	SPEAKER_01	Min læge på Regionshospitalet i Viborg henviste mig.
28.20s	32.10s	SPEAKER_00	Hvad hedder din læge?
32.10s	35.90s	SPEAKER_01	Han hedder Peter Holm.

Table 1: Example of a synthetically generated psychotherapy transcript illustrating the structure of the real data, including timestamps, speaker identifiers, and utterances. *Note:* The content is synthetic and does not originate from real patient data.

The example in Table 1 is synthetically gen-

erated using GPT-5.2 based on a predefined prompt shown in Table 2. A total of 100 rows were generated following the structure and label distribution of the real data (e.g., more PER and LOC entities and fewer MISC entities), mimicking the format and content of real psychotherapy conversations.

Prompt for Synthetic Transcript Generation
Generate 100 rows of synthetic psychotherapy transcript data following the format: start=<time> stop=<time> SPEAKER<ID> "text" The generated conversations should resemble Danish psychotherapy sessions and preserve a realistic distribution of named entity types (e.g., more PER and LOC entities and fewer MISC entities). [All entity categories from the annotation scheme described in Section 4.1 were included in the prompt.] Example rows: start=0.00s stop=2.34s SPEAKER_00 "Godmorgen, jeg hedder Lene Holm, og jeg er psykolog her." start=2.34s stop=5.91s SPEAKER_00 "Inden vi starter, vil jeg bare sige, at du er velkommen til at tage den tid, du har brug for." start=5.91s stop=9.88s SPEAKER_01 "Tak. Jeg kan mærke, at jeg er lidt nervøs for at sidde her." ...

Table 2: The 100 rows of synthetic transcripts were generated using GPT-5.2 based on this prompt.

The motivation for using synthetic data stems from practical constraints. Access to the real data was restricted to on-site use at the hospital (see Section 7.3), limiting the ability to develop and iteratively test the NER pipeline. As a result, the synthetic dataset was used as a workaround for remote prototyping, debugging, and model experimentation. Once the pipeline was error-free, all experiments were conducted on the real psychotherapy data.

3.3 Preprocessing

The preprocessing pipeline was first developed and tested on the synthetic data before being applied to the real psychotherapy transcripts. Starting from the .rttm raw transcription format, each utterance is transformed into a token-level representation suitable for named entity recognition (NER). Here, token-level refers to word-level tokens obtained using a rule-based tokenizer that separates punctuation from words.

The preprocessing pipeline is illustrated in Figure 2. First, starting from the .rttm transcription files, utterances are extracted together

with their associated speaker identifiers. Irrelevant audio segments, such as overlapping speech (speaker_OVERLAPPING) and noise (speaker_NOISE), are removed to ensure that only meaningful textual content is retained. Then, each utterance is tokenized using a rule-based tokenizer, separating punctuation from words. Throughout this process, relevant metadata such as session ID, speaker ID, and raw text is preserved. Finally, the tokenized output is converted into IOB format (Beginning-Inside-Outside), where each token is assigned a label indicating whether it begins an entity span (B-), continues an entity span (I-), or lies outside any entity (O).

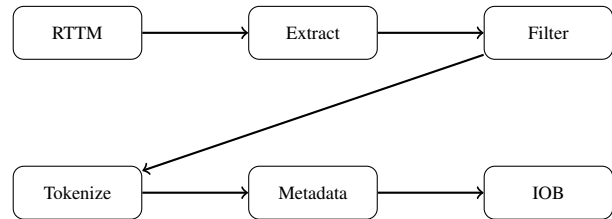


Figure 2: Preprocessing pipeline from .rttm transcripts to IOB format.

Token	Label
Godmorgen	O
,	O
jeg	O
hedder	O
Anne	B-PER
Sørensen	I-PER
,	O
og	O
jeg	O
er	O
psykolog	B-STAFF
her	O
på	O
klinikken	B-HOSPITAL
.	O

Table 3: Example of synthetically annotated data in IOB format. The same preprocessing pipeline is applied to both synthetic and real transcripts. *Note:* the name is not found in the real data.

For instance, the name "Anne Sørensen"

is labeled as B-PER (Anne) and I-PER (Sørensen), while "psykolog" and "klinikken" are annotated as STAFF and HOSPITAL, respectively. All remaining non-entity tokens are labeled O, as shown in Table 3 (Durango et al., 2023). The annotation scheme in this project includes domain-specific entity types such as HOSPITAL, STAFF, HEALTH, DIAGNOSIS, and MEDICATION, as defined in Section 4.1, in addition to general entity types such as PER, ORG, LOC, and MISC. This enables the model to detect not only explicit identifiers, but also context-dependent institutional and professional references relevant for anonymization.

The preprocessing pipeline transforms the transcripts into a structured representation compatible with standard NER pipelines. Each processed transcript contains a token index, a word token, and corresponding IOB labels. The preprocessing pipeline was first applied to the synthetic data to test for errors, and then to the real data.

3.4 Gold Dataset for Evaluation

To evaluate the performance of the NER models, we manually annotated one psychotherapy session from the real dataset using the defined entity tag scheme described in Section 4.1. The manual annotation process involved assigning IOB entity labels to tokens representing sensitive information, such as person names, locations, hospitals, diagnoses, and other domain-specific entities relevant for pseudonymization.

One psychotherapy session out of the 12 available sessions was randomly selected and annotated to serve as the gold dataset, i.e., the test dataset used for evaluation. The same preprocessing pipeline described in Section 3.3 was applied to the chosen psychotherapy session. After the annotation process, the gold dataset contained 100 B-tags and 39 I-tags. This gold dataset serves as the ground truth for model evaluation. By comparing the model

outputs to the manually annotated gold labels, we were able to compute evaluation metrics such as precision, recall, and F1-score to get a systematic assessment of each model’s ability to correctly identify sensitive entities in psychotherapy transcripts.

3.5 Inter-Annotator Agreement

To evaluate the quality of the manually annotated gold dataset, the inter-annotator agreement between the two annotators was measured. Assessing the annotation consistency in NER tasks is a common practice and is used in prior work, such as (Enevoldsen et al., 2024). Both annotators independently labeled the same psychotherapy transcript by following the guidelines described in Section 4.1. We report the agreement using both Cohen’s kappa coefficient and the token-level accuracy. Cohen’s kappa accounts for the agreement that occurs by chance, where p_o is the observed agreement and p_e is the expected agreement by chance. The formula is defined as:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

The annotators achieved a Cohen’s kappa score of 0.9620 and a token-level accuracy of 0.9994, indicating a high level of agreement. Only 10 tokens out of the entire transcript were labeled differently between the two annotators. To further assess agreement on the entity-level, the span-level F1-score was computed based on exact matches of both entity boundaries and labels. This resulted in an F1-score of 0.95, further supporting consistency in identifying entity spans. The transcript contained a relatively small number of entities, meaning that a few disagreements could significantly affect the agreement score. Most of the disagreements occurred in ambiguous cases, such as expressions of duration (e.g. "10 gange") or the use of English words (e.g. "Stray", "big community").

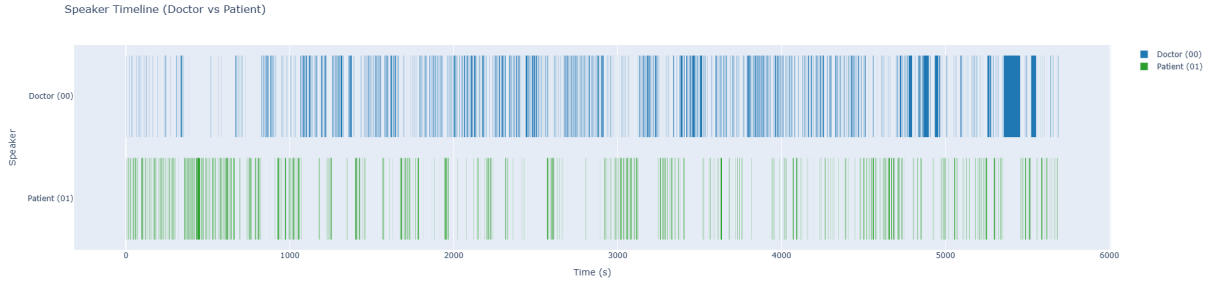


Figure 3: Speaker timeline showing therapist (blue) and patient (green) speech segments throughout the session.

In standard scenarios, agreement is evaluated iteratively to refine the guidelines and improve consistency. However, in our case, one iteration of agreement evaluation was sufficient based on the results. The high agreement scores suggest that the annotation guidelines were clear and consistently applied, making the gold data a reliable set for evaluating model performance.

3.6 Exploratory Data Analysis

We explored the speaker behavior and dialogue dynamics to better understand the structure and characteristics of the transcripts on the gold dataset. In particular, we examine how speaking time and utterances are distributed between the therapist and the patient, as well as how these patterns evolve over the course of a session (Figure 3).

The speaker timeline shows a dialogue structure with alternating turns between the therapist (blue) and the patient (green) throughout the session. The exchange is characterized by short contributions rather than long, uninterrupted monologues. However, certain segments, particularly later in the session, contain denser clusters of therapist speech, indicating periods where the therapist leads the conversation more actively. This is not surprising, as the therapist typically guides the discussion while the patient responds and reflects on their experiences.

The speaking time distribution further shows that the therapist speaks more overall

(Figure 4). The therapist accounts for approximately 59% of the total speaking time, while the patient accounts for around 41%. This difference is likely related to the therapist’s role in guiding the session and asking questions.

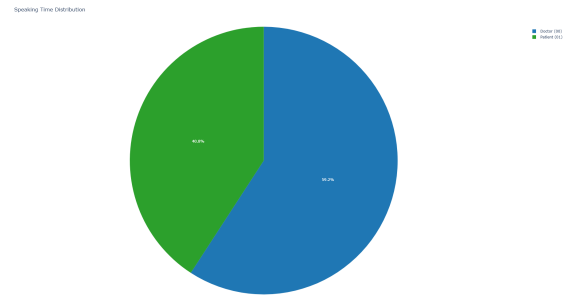


Figure 4: Distribution of speaking time between the patient (green) and the therapist (blue).

The distribution of utterances per speaker (Figure 5) shows that the therapist produces more utterances (1726) than the patient (1456), although the difference is not extreme. This suggests a balanced interaction, with both participants contributing to the conversation. The slightly higher number of therapist utterances again reflects the therapist’s role in asking questions, clarifying information, and steering the dialogue.

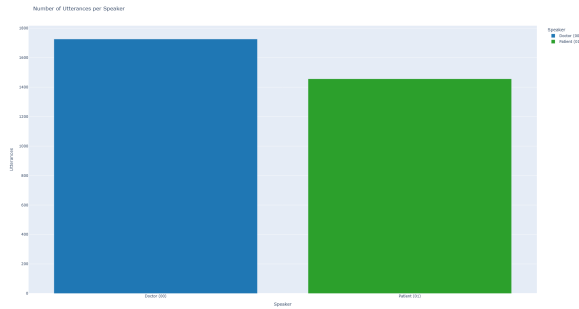


Figure 5: Number of utterances per speaker, where therapist utterances are shown in blue (00) and patient utterances in green (01).

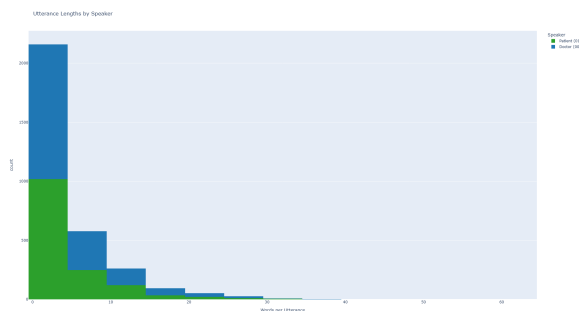


Figure 6: Distribution of utterance lengths per speaker measured in word tokens.

The distribution of utterance lengths reveals a strongly skewed pattern for both speakers (Figure 6), with the majority of utterances being short and a small number of longer utterances forming the tail of the distribution. The histogram bins represent utterance lengths measured in word tokens, grouping utterances by the number of words they contain (e.g., 0-4 words, 5-9 words, 10-14 words, and so on). As illustrated, mostly short sentences dominate the conversation. The therapist’s utterances seem to be slightly longer, whereas the patient’s utterances are shorter, often consisting of only a few words.

4 Methods

4.1 Annotation Scheme and Guidelines

Here, we present a proposed NER tag schema developed for the annotation and pseudonymization of psychotherapy transcripts in our project. The purpose of the

schema is to systematically identify personal and sensitive information in accordance with GDPR. Specifically, the schema covers personal data under GDPR, including direct identifiers (e.g., names, addresses), indirect identifiers (e.g. relationships, locations), and sensitive data such as health and mental health information relevant to psychotherapy.

The initial set of entity categories was derived from the Text Anonymization Benchmark (TAB) (Pilán et al., 2022), which provides an established framework for identifying personal information in text. These categories were subsequently refined to better reflect the characteristics of psychotherapy data, where sensitive contextual details about individuals frequently occur.

To ensure legal relevance, each tag was evaluated using GDPR definitions. GDPR Article 4 (European Parliament and Council of the European Union, 2016b) was used to determine which types of information may directly or indirectly identify an individual, while GDPR Article 9 (European Parliament and Council of the European Union, 2016a) guided the inclusion of special categories of personal data such as health information, ethnicity, religion, political opinions, and sexual orientation.

The resulting tag set, therefore, combines an established NLP anonymization framework with GDPR-based justification, supporting consistent annotation and transparent handling of identifiable and sensitive information.

Parent tags	Child tags (and what they cover)	GDPR Article Reference
PER	PER, STAFF, PATIENT, RELATION Names of people, including nicknames/aliases, usernames, initials, and other people mentioned such as family members (relation).	Art. 4 GDPR Definitions (1): Personal data is any information relating to an identified or identifiable natural person. Names directly identify a person.
LOC	LOC, ADDRESS, POSTCODE Places and locations such as cities, areas, countries, addresses, infrastructures, departments, streets, ZIP codes, rooms etc.	Art. 4 GDPR Definitions (1): Location data can make a person identifiable directly or indirectly.
ORG	ORG, HOSPITAL Names of organizations such as public and private companies, schools, universities, public institutions, prisons, healthcare institutions, NGOs, churches etc.	Art. 4 GDPR Definitions (1): Organizational affiliation can identify a natural person when combined with other information.
CONTACT	CONTACT, PHONE, EMAIL Emails, telephone numbers, fax numbers and other contact information.	Art. 4 GDPR Definitions (1): Contact details relate to an identifiable individual.
ID	ID, CPR Identification codes such as social security numbers (CPR), passport numbers, license plates etc.	Art. 4 GDPR Definitions (1): Identification numbers are explicitly listed as personal data.
DATE	DATE, TIME, DURATION Descriptions of specific dates (e.g., October 3 2018), times (e.g., 9:48 AM), or durations (e.g., 18 years).	Art. 4 GDPR Definitions (1): Dates become personal data when linked to an identifiable person.
DEM	DEM, ETHNICITY, RELIGION, POLITICS, SEXUALITY, AGE Demographic attributes such as ethnicity, religion, political opinions, sexuality, job titles, education, physical descriptions, and age.	Art. 9 GDPR Processing of special categories of personal data (1): Includes racial or ethnic origin, political opinions, religious beliefs, sex life or sexual orientation.
HEALTH	HEALTH, DIAGNOSIS, MEDICATION, CONDITION Personal data related to physical or mental health including diagnoses, medical conditions, substance use and genetic predispositions mentioned explicitly or implicitly in text.	Art. 9 GDPR Processing of special categories of personal data (1): Data concerning health is explicitly classified as a special category of personal data.
MISC	No child tag. Every other type of personal information associated directly or indirectly with an individual and not belonging to the categories above.	Art. 4 GDPR Definitions (1): Any information that can identify a person directly or indirectly qualifies as personal data.

Table 4: NER tag categories and their relation to GDPR personal data definitions

If an entity cannot be confidently assigned to any available child tag, it will be annotated using the corresponding parent tag. Annotators are instructed to apply the most specific label when possible, but to default to the parent category in cases of uncertainty or insufficient granularity in the child taxonomy. The hierarchical structure of parent and child tags also serves as a rollback mechanism for the anonymization pipeline. Should evaluation show that fine-grained child tags reduce annotation consistency or anonymization performance, the annotations can be mapped back to the broader parent-level categories without requiring re-annotation of the dataset.

4.1.1 Identifiers

Identifiers in the dataset are divided into direct identifiers and quasi-identifiers. Direct identifiers refer to attributes that can uniquely identify an individual on their own, such as names, CPR numbers, contact information, or addresses. These identifiers must therefore be anonymized to prevent re-identification. In contrast, quasi-identifiers do not directly reveal an individual’s identity when considered in isolation, but may enable re-identification when combined with other information (Pilán et al., 2022). Table 5 provides an overview of how the entity types used in this project are categorized into direct identifiers and quasi-identifiers.

Direct Identifiers	Quasi-Identifiers
PER (incl. all child tags)	ORG (incl. all child tags)
ID (incl. all child tags)	DATE (incl. all child tags)
CONTACT (incl. all child tags)	LOC, POSTCODE
ADDRESS (child of LOC)	DEM (incl. all child tags)
	HEALTH (incl. all child tags)
	MISC

Table 5: Classification of entity tags into direct and quasi-identifiers.

4.2 Off-the-shelf NER model

We evaluate two off-the-shelf pretrained NER systems from spaCy as baseline models, representing a small and a large Danish news

pipeline (spaCy, 2026a). For the off-the-shelf baseline model, we used *da_core_news_sm*, spaCy’s small Danish news pipeline, and *da_core_news_lg*, spaCy’s large Danish news pipeline (spaCy, 2026a).. These are pre-trained models trained on Danish news text and include components for tokenization, part-of-speech (POS) tagging, dependency parsing, and named entity recognition (NER). It consists of a tok2vec embedding layer with 500,000 pretrained 300-dimensional word vectors (spaCy, 2026a). The NER component predicts a standard set of coarse-grained entity types such as PER (person), ORG (organization), LOC (location), and MISC (miscellaneous) (spaCy, 2026a). The small version of the model (sm) does not include large word vectors, and it is designed to be lightweight and fast rather than highly specialized (spaCy, 2026b). It therefore serves as a reasonable general domain baseline, but it is not tailored to psychotherapy transcripts or sensitive health-related data, missing a lot of entities in the text. The large version of the model (lg) performed a bit better, because it includes pretrained word vectors (spaCy, 2026b), which help the model capture more semantic information about words and their context (Mikolov et al., 2013).

To evaluate the models in our setting, we applied them to our generated psychotherapy transcripts and the annotated gold data. The data was formatted in the IOB2 tagging scheme, where each token is assigned a "B-", "I-", or "O" label to mark entity boundaries explicitly at the token level. Since our gold annotations use fine-grained label set (e.g. HEALTH, DIAGNOSIS, AGE, CPR; for more information refer to Table 4), while the spaCy model predicts a smaller and more generic entity types, we mapped the gold labels into a more coarse label space (PER, ORG, LOC, MISC) to enable direct comparison, for example mapping PER, STAFF, PATIENT -> PER,

LOC, ADDRESS, POSTCODE -> LOC, ORG, HOSPITAL -> ORG and rest to MISC. The spaCy predictions were converted into IOB format and aligned with the gold tokens, after which precision, recall, and F1-score were computed using standard sequence evaluation metrics. This setup allows us to assess how a general domain, off-the-shelf Danish NER models perform on psychotherapy style data without any domain-specific fine-tuning.

4.3 Commercial EU PII Detection System

4.3.1 Model Description

For the commercial baseline, we used EU PII Safeguard, a multilingual token-classification model for personally identifiable information (PII) detection developed by Tabularis AI (Tabularis AI). The model is available on Hugging Face and is described as a GDPR-oriented system for multilingual PII detection in European languages. It is based on XLM-RoBERTa-large and contains approximately 0.6 billion parameters. The underlying XLM-RoBERTa-large model is pretrained on large-scale multilingual corpora using a masked language modeling objective (Conneau et al., 2019a). The PII detection model is fine-tuned for token classification, although detailed information about the specific training data and annotation process used by Tabularis AI is not publicly disclosed.

According to the model documentation, EU PII Safeguard supports 26 European languages and predicts BIO-formatted labels corresponding to 42 entity types (74 labels including B- and I- prefixes). These entity types cover a wide range of PII categories, including personal attributes, contact information, identity, financial data, health-related information, and technical identifiers.

Category	Examples of entity types
Health	health condition, health insurance ID
Names	first name, last name, middle name, prefix
Personal attributes	age, date of birth, gender, ethnicity, religion, sexual orientation, political opinion, job title
Contact	address, street, city, country, postal code, email, phone number
Identity	national ID, passport number, driver license, tax ID
Financial	account number, credit card number, IBAN, salary, amount, currency
Others	company name, username, password, URL, IP address, MAC address, device ID, building number, state, latitude, longitude

Table 6: Overview of the entity categories supported by EU PII Safeguard.

The model supports a maximum input length of 256 tokens, which limits the amount of context that can be processed in a single forward pass.

In this project, the model was used in a zero-shot setting. No fine-tuning was performed on psychotherapy transcripts or on the manually annotated gold data. The model is therefore treated as a commercial model baseline.

4.3.2 Model Configurations

We evaluated two different model configurations for handling context.

First, we applied a sentence-by-sentence setup, where each sentence was processed independently. This represents a minimal-context baseline.

Second, we evaluated a sliding-window approach with window sizes of 3, 5, 7, and 9 sentences. In this setup, each sentence was processed together with its surrounding context. This design was motivated by the nature of psychotherapy transcripts, where identifiers may depend on information expressed in neighboring sentences rather than on a single sentence alone.

4.3.3 Evaluation Data

The commercial model was evaluated on the main evaluation dataset, which was shared across all participants. This dataset was used for all model comparisons and for reporting the main evaluation results.

4.3.4 Selection of the Final Configuration

The sentence-based and sliding-window configurations were compared on the main evaluation dataset. The configuration with the best performance was selected as the final setup. This selected configuration was then used for all reported results of the commercial model.

4.3.5 Prediction Alignment and Label Mapping

The model produces token-level BIO predictions. To enable comparison with our annotated data, predictions were aligned to the original IOB2 tokenization.

Since the label set of EU PII Safeguard differs from our annotation scheme, predicted labels were mapped to a shared evaluation label space. Table 7 presents the most relevant subset of this mapping; the complete mapping is provided in Appendix A.

Parent tag	Mapped EU PII Safeguard labels
PER	first name, last name, middle name, username, prefix
LOC	address, street, city, country, postal code, state, building number, latitude, longitude
ORG	company name
CONTACT	email, phone number
ID	account number, credit card number, IBAN, national ID, passport number, driver license, tax ID
DATE	date of birth
DEM	age, gender, ethnicity, religion, sexual orientation, political opinion, job title
HEALTH	health condition
MISC	amount, currency, salary, password, URL, IP address, MAC address, device ID

Table 7: Subset of the mapping from EU PII Safeguard labels to the parent-tag evaluation space. BIO prefixes (B-, I-) were preserved during mapping.

4.3.6 Evaluation Protocol

Evaluation was performed at the entity level using seqeval. Precision, recall, and F1-score were computed by comparing predicted entity spans to the gold annotations. The same evaluation procedure was used across all experimental settings.

4.4 Few-shot prompting approaches

4.4.1 Model selection

For the few shots approach, we at first chose and initialized the attempt with the Mistral-7B-instruct-v0.2 model. This model is a 7-billion-parameter instruction-tuned large language model developed by Mistral AI (Mistral AI). It is designed to follow structured prompts and generate controlled outputs, which makes it a suitable option for our case. The model includes a transformer-based decoder architecture (Vaswani et al., 2017) and supports context windows of up to 8000 tokens, which allows it to process relatively long prompts, including instructions, describing what the model should do with the input it is given and few-shot examples. We put

the model to use through a cloud-based environment, Databricks, since sufficient GPU resources were needed and required to run the model on the dataset. However, after preliminary experimentation, it became clear that the chosen model is not capable of performing the given task. The model is described as multilingual, meaning it has been trained on text from multiple languages, but in our experiments, its performance on the Danish language seemed to be insufficient for the task of named entity recognition. We did not manage to obtain any reasonable results using this model, and as a result, we searched for an alternative model better suited for our case and data. This resulted in the choice of `snakmodel-7b-instruct` (NLP-north), which is a Danish-focused large language model developed by NLPnorth. It is based on a transformer architecture (Vaswani et al., 2017) and is likewise instruction-tuned. It has approximately 7 billion parameters, similar to the previously used model. Long input context is supported, allowing detailed task descriptions and examples for the few-shot prompting. Unlike the previous model, SnakModel is trained primarily on Danish data, which allows a better grasp of understanding vocabulary and entity boundaries in Danish text.

4.4.2 Model configuration

We run the SnakModel in a Databricks environment, along with an instruction prompt that was set, describing a restricted set of allowed entity labels (see Table 4) and a description of what the model is expected to do with the given input. The model processed the dataset sequentially, where each sentence was provided as a separate input instance. Furthermore, the model was instructed to strictly output predefined JSON format labeled sentences, e.g., “entities”: [{"surface": "2019", "label": "DATE"}], which helped reduce inconsistency in the responses that were generated. Instead of outputting long explanatory statements, the model

was constrained to produce a fixed and structured output containing entity spans and corresponding labels. Generation was also limited by setting a maximum number of new tokens to 40, and a zero-temperature setting to ensure more consistent and sensible output.

4.4.3 Output processing

Post-processing was included for the generated outputs of the model in order to obtain consistency throughout the output as a whole. Since the model was instructed to generate JSON format answers, the first step was to validate and clean the obtained output, so valid JSON structures could be extracted. If the output contained an invalid format or had additional text attached, a normalization function was included to recover the content if possible. If no valid entities were detected or the output could not be interpreted, an empty entity list was returned instead.

4.4.4 Evaluation process

Since the outputs of the model were generated in a JSON format, they were first converted into the IOB2 schema for further comparison. Additionally, our evaluation data contained a finer-grained set of labels than those used by the model during inference. Therefore, the evaluation data were remapped to correspond to the label set used by the SnakModel, so that predictions and evaluation labels were directly comparable. The final evaluation itself was conducted using the `sequeval` (Nakayama) library, which computes token-level and entity-level metrics on the matching IOB2 spans.

Original evaluation label	Remapped model label
STAFF	PER
HOSPITAL	ORG
ADDRESS	CONTACT
TIME	DATE
DIAGNOSIS	HEALTH
RELIGION	MISC

Table 8: Label remapping from the evaluation data to the SnakModel label set.

4.5 Fine-tuned transformer-based model

4.5.1 Model Choice

We utilized XLM-RoBERTa-base and the bigger variant -large as the underlying models for the named entity recognition (NER) task. XLM-RoBERTa is a multilingual transformer model pretrained on 2.5 TB of large-scale CommonCrawl data covering 100 different languages (Conneau et al., 2019b). One of the languages present in the corpus is Danish, with approximately 7.8 billion tokens making up 45.6 GiB, which provides a reasonable foundation for learning linguistic patterns in Danish. Other linguistically similar languages, such as Norwegian and Swedish are also included in similar amounts, which may further support the learning of shared lexical patterns found across these languages. Although the task is focused on Danish psychotherapy transcripts, the use of a multilingual model is motivated by its ability to learn transferable representations across languages. Prior work suggests that multilingual pretraining may improve generalization in low-resource and domain-specific settings where labeled data is limited (Conneau et al., 2019b). Therefore, this could be relevant in the clinical domain, where annotated Danish data is scarce.

The models are designed to be fine-tuned for token classification tasks such as NER, which is suitable for identifying sensitive information in text. By using the pretrained contextual representations, the models can be adapted to the domain-specific task while keeping general linguistic knowledge. Alternatively, a Danish-specific model could be used; however, such a model may lack the generalization abilities provided by multilingual pre-training (reference needed). Given the conversational nature of psychotherapy data, the broad linguistic coverage of XLM-RoBERTa may be beneficial.

Category	DaCy Entity Types
Person-related	PERSON
Location-related	LOCATION, GPE, FACILITY
Organization-related	ORGANIZATION
Temporal	DATE, TIME, EVENT
Demographic	NORP, LANGUAGE, LAW, WORK OF ART
Numerical	CARDINAL, ORDINAL, MONEY, PERCENT, QUANTITY
Other	PRODUCT

Table 9: Overview of the 18 entity types used in the DaCy fine-grained NER model, grouped by semantic category.

4.5.2 Fine-tuning setup

The pretrained XLM-RoBERTa models are fine-tuned on the Danish psychotherapy data annotated in the IOB2 format. Each sentence is tokenized using the XLM-RoBERTa tokenizer, which is based on SentencePiece subword segmentation (Kudo et al., 2025). The resulting subword tokens are aligned with their corresponding labels before being passed to the models. The task itself is formulated as a token classification, where the models predict a label distribution for each token in the sequence. The training is performed using the HuggingFace Transformers library. Cross-entropy loss is computed over tokens, and model parameters are updated through backpropagation. The label space is constructed dynamically from the training and evaluation data, which introduces a limitation in that the model can only predict labels observed during fine-tuning.

The transcripts used for fine-tuning were pre-annotated using the chcaa/da_dacy_large_ner_fine_grained model, which is part of the DaCy framework (Enevoldsen et al., 2024). DaCy provides a Danish NER pipeline based on the DANSK dataset containing 18 entity types from the OntoNotes annotation scheme as summarized in Table 9. To adapt the data for our task, these 18 entity types were mapped to our annotation scheme shown in (4). While some labels (e.g. PER, LOC, ORG, DATE) map directly, others were assigned based on semantic similarity. For instance, LANGUAGE and NORP (nationalities, political groups etc.)

were mapped to DEM. Numerical categories such as CARDINAL, ORDINAL, MONEY, and PERCENT were mapped to MISC, as they did not correspond directly to sensitive entity types relevant for pseudonymization. This remapping introduces some trade-offs. On one hand, it allows us to retain more training data in a setting with few annotated entities. On the other hand, it also introduces noise in the label distribution. For example, mapping several labels to MISC may encourage the model to overgeneralize as there is much diversity in the MISC label.

The dataset of 12 sessions was split into a 10/1/1 distribution for train, evaluation, and test set. Across these 12 sessions, there were approximately 1400 annotated entities. The amount of annotated entities can be a potential limitation, as it may restrict the models’ ability to learn representations for less frequent entity types. Finally, the other limitation is that the labels originate from a separate NER model (DaCy) and are subsequently remapped; annotation errors may propagate into the fine-tuning process. Ideally, the data should be manually annotated according to the target label scheme to ensure higher quality and consistency.

4.5.3 Hyperparameters

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1×10^{-5} (Base, Large) 1×10^{-4} (LoRA)
Batch size (train)	2
Batch size (eval)	2
Grad. Accum.	8
Epochs	3
Weight decay	0.01

Table 10: Training hyperparameters for XLM-RoBERTa models

The hyperparameters used in the fine-tuning process are shown in Table 10. All models were trained using the AdamW optimizer. A

lower learning rate of 1×10^{-5} was used for the base and large model variants to ensure stable updates of pretrained weights. In contrast, a higher learning rate of 1×10^{-4} is applied for the LoRA-adapted large model, since only a small subset of parameters (0.4%) is updated during training. Otherwise, using a lower learning rate for the LoRA setup resulted in unwanted model behavior, where the model predicted only the non-entity label O, indicating problems in the training phase. The batch size was set to 2 for both the train and evaluation sets due to memory limitations encountered on the Databricks cluster. To compensate for the small batch size, gradient accumulation is applied over 8 steps. The models are trained for 3 epochs, which was considered balanced between learning and overfitting given the limited dataset size. A weight decay of 0.01 is applied as regularization to prevent any overfitting by penalizing large parameter values.

All hyperparameters, except the learning rate used for the LoRA setup, were fixed across experiments to reduce the number of experimental variables and simplify comparison between model variants. However, it should be noted that optimal hyperparameter configurations may differ across architectures, meaning that the reported performance differences cannot be solely attributed to architectural differences. For instance, larger transformer models and approaches such as LoRA may respond differently to learning rate, batch sizes and other regularization hyperparameters (Hu et al., 2021). Ideally, an extensive hyperparameter optimization procedure such as grid search, could potentially yield improvements to the individual models. However, this was considered outside the scope of the project.

4.5.4 Model Architecture and Training

The tokenized input goes through multiple transformer layers, which output contextualized embeddings for each token. A linear classification head is added on top of the encoder

Model	# Params	Trainable	Layers	Train Time
XLM-R Base	270M	270M	12	1h 42m
XLM-R Large	550M	550M	24	2h 36m
XLM-R Large + LoRA	561M	2.38M (0.4%)	24	1h 17m

Table 11: Comparison of model architectures, trainable parameters, and training time. Architecture details are based on (Conneau et al., 2019b).

to predict a label distribution for each token in the sequence. Table 11 shows how the base and large variants each differ in architecture with 12 and 24 transformer layers and 270M and 550M parameters, respectively (Conneau et al., 2019b). Including both the small and large variants gives a better representation of the trade-off between model size and computational efficiency. The large model is expected to achieve better performance due to increased depth and parameters, while the base model provides shorter training time. Finally, the LoRA-adapted model keeps the architecture of the large model, but updates happen only on a small fraction (0.4%) of its parameters, resulting in the fastest training time, as shown in Table 11.

4.5.5 LoRA Adaptation

LoRA Hyperparameter	Value
Rank r	16
LoRA α	32
LoRA Dropout	0.1
Bias	none
Target modules	query, key, value

Table 12: LoRA configuration for XLM-RoBERTa Large

To improve parameter efficiency, we experiment with Low-Rank Adaptation (LoRA) applied to the XLM-RoBERTa-large model (Hu et al., 2021). LoRA is a parameter-efficient technique that avoids updating the whole set of pretrained model weights. Instead, it introduces low-rank matrices into selected parts of the transformer architecture, while keeping the original pretrained weights frozen. In our

setup, LoRA is applied to the attention mechanism by injecting low-rank matrices into the query, key, and value projections as shown in Table 12. During the training phase, only approximately 0.4% of all parameters are trained for the large model, corresponding to 2.38M parameters out of 561M. This led to a reduced memory usage and faster training as illustrated in Table 11. The rest of the LoRA configuration used is shown in Table 12. The rank r controls the dimensionality of the adaptation, while the scaling factor α adjusts the magnitude of the updates. A dropout of 0.1 is applied, which can improve generalization. The choice of targeting the query, key, and value matrices follows common practice, as these components play a central role in the attention mechanism.

Overall, by restricting the number of trainable parameters, LoRA enabled us to experiment faster and use less computational resources in comparison to a full fine-tuning of large transformer models.

4.6 Anonymization Pipeline

The anonymization in our project is implemented as a pipeline that combines entity detection with rule-based pseudonymization. At a high level, the process consists of two stages. First, a NER model is used to detect sensitive entities in the transcript. Second, the detected entity spans are transformed into pseudonymized replacements. This setup follows the standard anonymization workflow, in which sensitive information is first identified and then transformed through modification, removal, or replacement (Vakili et al., 2023).

Common anonymization strategies include replacing an entity with its category label, a surrogate value, or a category-specific placeholder. The latter two strategies are both used in our project. Person-related entities such as PER, PATIENT, and STAFF are replaced with Danish surrogate values, while broader categories such as MISC, DEM, and HEALTH are

replaced with generic placeholders such as [diverse info], [demografisk info], and [helbredsoplysninger]. Temporal expressions such as dates and times are handled through shifted surrogate values when possible. This corresponds to a pseudonymization-oriented approach, where sensitive expressions are replaced with alternatives of the same type rather than being fully removed (Vakili et al., 2023). In our project, we adopt this approach because the goal is not only privacy protection, but also the preservation of linguistic plausibility in transcripts intended for later LLM-related use. Replacing a name with another realistic Danish name, or a location with a Danish placeholder such as [dansk by], preserves the structure of the text better than full redaction and is therefore more suitable for downstream modeling (Yermilov et al., 2023).

The pipeline operates on token-level predictions in IOB2 format. Tokens labeled as O are preserved unchanged. Tokens labeled with B- and I- prefixes are grouped into contiguous entity spans according to their tag type. This step is essential because many sensitive expressions in psychotherapy transcripts are multi-token units, such as dates, durations, person names, or institution names. Instead of pseudonymizing each token separately, the pipeline first reconstructs the full entity span and then replaces the span as a single unit. This avoids malformed output in cases where only the first token would otherwise be replaced.

After reconstructing a full entity span, the pipeline applies a tag-specific replacement function. The replacement strategy is not uniform across entity types. Instead, each tag is associated with a pseudonymization rule depending on whether the goal is to preserve plausibility, preserve format, or remove sensitive information more aggressively. In general, person-, location-, and institution-related entities are replaced with surrogate values, identifier-like entities are replaced using format-preserving

Tag	Replacement	Strategy
PER, PATIENT	Surrogate	Random Danish first name
STAFF	Surrogate	Role-bearing form Psykolog <Name>
RELATION	Surrogate	Family relation from predefined list
LOC	Placeholder	Fixed placeholder [dansk by]
ADDRESS	Placeholder	[dansk adresse]
ORG	Placeholder	[organisation]
HOSPITAL	Placeholder	Fixed placeholder [dansk hospital]
PHONE, EMAIL	Format-preserving	Random values matching valid format
ID, CPR	Format-preserving	Random 10-digit identifiers
DATE, TIME	Shifted surrogate	Shifted relative to original if parsable, otherwise synthetic
DEM, HEALTH, MISC	Placeholder	Fixed placeholders (e.g. [helbredsoplysninger], [diverse info])

Table 13: Overview of pseudonymization strategies used in the anonymization pipeline. Full mapping is provided in Appendix B.

values, and broad sensitive categories such as DEM, HEALTH, and MISC are replaced with generic placeholders. Temporal expressions such as dates and times are handled separately through shifted replacements in order to preserve a more natural textual form.

The anonymization pipeline, therefore, acts as the final transformation layer between entity detection and usable pseudonymized output. This distinction is important because accurate entity detection alone does not automatically produce a privacy-preserving transcript. The practical usefulness of the system depends on how predicted spans are reconstructed, replaced, and constructed back into the text.

5 Results

5.1 Off-the-shelf NER model

5.1.1 Results - small model

Label	Precision	Recall	F1-score
LOC	0.37	0.65	0.47
MISC	0.40	0.07	0.12
ORG	0.10	0.25	0.14
PER	0.12	0.55	0.19
Macro avg	0.25	0.27	0.21
Micro avg	0.22	0.27	0.24

Table 14: Performance of *da_core_news_sm* on gold data

This result shows that the small model performs poorly on the psychotherapy session data with varying performance across entities. The recall for LOC (0.65) and PER (0.55) is average, indicating it detects many true entities, but with low precision (0.37 for LOC, 0.12 for PER), meaning many false positives. Performance for ORG and MISC is particularly weak, with extremely low recall for MISC (0.07), indicating that the model fails to capture less common entity types, such as the many quasi-identifiers we have defined for us. Overall, this result is not satisfying and therefore *da_core_news_sm* is not suitable for this highly domain-specific task of psychotherapy session transcripts, as it fails to detect both direct and quasi-identifiers.

5.1.2 Results - large model

Label	Precision	Recall	F1-score
LOC	0.54	0.65	0.59
MISC	0.62	0.09	0.16
ORG	0.08	0.25	0.12
PER	0.27	0.91	0.42
Macro avg	0.38	0.48	0.32
Micro avg	0.36	0.33	0.34

Table 15: Performance of *da_core_news_lg* on gold data

This result shows a small improvement over the small model. The most notable result is the high recall for PER (0.91) combined with low precision (0.27), indicating that the model aggressively labels tokens as PER, resulting in many false positives (low precision). Performance of LOC is mediocre (recall 0.65, precision 0.54), however ORG and MISC remain weak. Overall, the model captures some entities effectively, but its tendency to overpredict PER and underpredict MISC and ORG makes it unsuitable for this kind of task.

5.2 Commercial PII Detection System

5.2.1 Comparison of Model Configurations

We first compared different model configurations of the commercial model on the main evaluation dataset, including a sentence-by-sentence setup and sliding-window approaches with window sizes of 3, 5, 7, and 9 sentences.

Configuration	P	R	F1
Sent-by-sent	0.387	0.120	0.183
Window-3	0.345	0.200	0.253
Window-5	0.290	0.200	0.237
Window-7	0.358	0.240	0.287
Window-9	0.333	0.230	0.272

Table 16: Micro-averaged precision, recall, and F1 per configuration of EU PII Safeguard (zero-shot). Best in bold.

The sliding-window approaches consistently outperformed the sentence-by-sentence baseline, indicating that the surrounding context improves entity detection in psychotherapy transcripts. The sentence-by-sentence setup achieved the highest precision (0.387) but at the cost of very low recall (0.120), suggesting that without context, the model is overly conservative and misses the majority of entities. Among the evaluated configurations, the **window-7** setup achieved the best overall F1-score (0.287) and was therefore selected as the final configuration.

5.2.2 Results on the Main Evaluation

Dataset

The selected configuration was evaluated on the main evaluation dataset. Results are shown in Table 17. Scores reflect performance after label mapping (see Table 7).

Label	Precision	Recall	F1-score
LOC	0.524	0.550	0.537
PER	0.320	0.727	0.444
DEM	0.267	0.174	0.211
HEALTH	0.167	0.077	0.105
Macro avg	0.160	0.191	0.162
Micro avg	0.358	0.240	0.287

Table 17: Per-label performance of EU PII Safeguard on the main evaluation dataset (window size 7, zero-shot), after mapping to the shared evaluation label space.

The model achieves a micro F1-score of 0.287, with LOC (F1 = 0.537) and PER (F1 = 0.444) as the strongest labels. This is expected, as these categories map most directly from the model’s native label space: city, country, street for LOC and first name, last name for PER.

The zero scores for DATE, ID, MISC, and ORG are explained by the label mapping. DATE in our gold data consists of conversational expressions such as *sidste weekend*, *halvanden time*, and *20 minutter frem*, while the mapping only covers *date of birth*. ORG maps solely from *company name*, which does not match informal references like *rådhuset* or *museerne*. ID and MISC map from formal document numbers and financial tokens respectively, neither of which appear in psychotherapy speech.

DEM and HEALTH both score low. For DEM, the mapping covers age, gender, ethnicity, and job title among others, yet recall remains at 0.174. This suggests that demographic attributes in conversational speech, often mentioned implicitly or mid-sentence, are

harder for the model to detect than in a more structured text. HEALTH suffers similarly, with recall of only 0.077; clinical terms such as *angst*, *psykotisk*, and *terapi* are frequently missed despite *health condition* being part of the mapping.

For PER, the high recall (0.727) comes at the cost of low precision (0.320), meaning the model catches most person names but also tags many non-entities as PER. This pattern is discussed further in the qualitative analysis.

5.3 Few-shot prompting

5.3.1 Results of Mistral-7B-instruct-v0.2

Label	Precision	Recall	F1-score
Micro avg	0.02	0.14	0.04
Macro avg	0.02	0.02	0.01

Table 18: Result metrics for Mistral-7B-Instruct-v0.2 on the evaluation dataset

The evaluation results for the Mistral-7B-instruct-v0.2 model (Table 18) showed that the model has achieved overall low performance on the task. Since most of the individual entity labels resulted in zero precision, recall, and F1-scores, Table 18 presents only the aggregated evaluation metrics. The model achieved a micro-average precision of 0.02, a recall of 0.14, and an F1-score of 0.04. Likewise, the macro-averaged evaluation resulted with precision and recall both at 0.02 and an F1-score of 0.01, reflecting that the model was not capable to reliably detect entities across individual categories. The weighted-average F1-score reached 0.10. These results suggest that the model managed to identify only a limited portion of the targeted entities and also produced a large number of incorrect predictions.

5.3.2 Results of snakmodel-7b-instruct

The SnakModel evaluation (Table 19) showed a noticeable improvement over the initial experiment. The model resulted in a micro-averaged precision of 0.03, a recall of 0.32, and

Label	Precision	Recall	F1-score
CONTACT	0.000	0.000	0.000
DATE	0.154	0.154	0.154
DEM	0.105	0.087	0.095
HEALTH	0.200	0.154	0.174
ID	0.000	0.000	0.000
LOC	0.149	0.700	0.246
MISC	0.000	0.000	0.000
ORG	0.125	0.250	0.167
PER	0.081	1.000	0.150
Macro avg	0.090	0.261	0.109
Micro avg	0.033	0.320	0.060

Table 19: Result metrics for SnakModel-7B-Instruct on the evaluation dataset.

an F1-score of 0.06, with a macro-averaged F1-score of 0.11 and a weighted-average F1-score of 0.14. It can be clarified that the difference between recall and precision suggests that while the model managed to capture a reasonable amount of the relevant entities, it at the same time produced a considerable number of false positives.

The performance of the model varied to a large extent from one entity category to another. The PER label achieved a recall of 1.00, which indicates that the label was identified consistently but at a cost of low precision. The LOC category also performed relatively well and resulted in a recall of 0.70 and an F1-score of 0.25, which suggests it was one of the best-performing labels for this setup. Categories such as HEALTH, ORG, and DATE returned moderate results, whereas CONTACT, ID, and MISC were largely undetected in the evaluated samples.

Another observation was that the model struggled frequently when the input instance contained a longer or more complex sentence. In such cases, the predicted entity labels in the generated output often broke the predefined JSON structure, which resulted in malformed outputs and reduced the ability of label extraction.

Although the SnakModel results show that

the model was capable of recognizing certain entity types across Danish psychotherapy transcripts, it also shows its extraction quality remained uneven across different label categories. It is also meaningful to note that the final F1-scores were limited by the low precision values, which suggest that incorrect predictions were made frequently by the model.

5.4 Fine-tuned transformer-based model

5.4.1 Results of XLM-RoBERTa model variants

Table 20 presents the performance of the XLM-RoBERTa variants across entity types. In general, the results show moderate performance with variation across entity types, reflecting the limitations of a small dataset and label remapping. The first observation is that PER and LOC are the best-performing entity types across all model variants. The large model achieves the highest performance on PER (F1 = 0.69), mainly due to a strong recall of 0.82, meaning that the large model is effective at identifying person entities. In a similar manner, LOC achieves reasonable performance across models (F1 = 0.61–0.65), suggesting that location-related entities are learnable for the models to some degree. On the other hand, DEM and DATE show imbalanced precision-recall trade-offs. For instance, in the large model, DEM shows a high precision (0.83), but low recall (0.22). This may be due to the heterogeneous content of the DEM category, which includes a lot of varied concepts such as nationality, language, and political affiliation after remapping. Also, DATE achieves low precision but a slightly higher recall, suggesting that the models predict it more but not as accurately. The MISC category performs consistently poorly across all models (F1 ≤ 0.11). This is expected, as MISC includes a mixture of remapped entities (e.g. different numerical and other types), making it a noisy and difficult to predict correctly. Therefore, merging

Label	XLM-R Base			XLM-R Large			XLM-R Large + LoRA		
	P	R	F1	P	R	F1	P	R	F1
PER	0.64	0.64	0.64	0.60	0.82	0.69	0.78	0.64	0.70
LOC	0.64	0.58	0.61	0.74	0.58	0.65	0.70	0.58	0.64
DEM	0.83	0.22	0.34	0.83	0.22	0.34	0.67	0.17	0.28
DATE	0.11	0.31	0.16	0.10	0.31	0.15	0.10	0.38	0.16
MISC	0.06	0.10	0.08	0.06	0.10	0.08	0.10	0.14	0.11
Macro avg	0.46	0.37	0.37	0.47	0.41	0.38	0.47	0.38	0.38
Micro avg	0.27	0.33	0.30	0.27	0.35	0.30	0.27	0.34	0.30

Table 20: Comparison of NER performance across XLM-RoBERTa models on gold data. Micro and macro averages are included to summarize overall performance across entity types.

several entity types into one is not ideal, as it introduces ambiguity, resulting in bad performance. On the whole, the large model outperforms the base model, particularly for PER and LOC, showing that an increased model size captures contextual patterns better. The LoRA-adapted model has competitive performance despite updating only a small fraction (0.4%) of parameters. It achieves the highest precision for PER (0.78), but with a lower recall score compared to the full fine-tuned large model. For the aggregated metrics, micro-averaged F1-scores are identical for all models (0.30), however macro-averaged F1-scores slightly favor the larger models (0.38 vs. 0.37).

Finally, it is important to note that not all entity types from our label scheme are present in the results. Specifically, categories such as CONTACT, ID, and HEALTH are absent. This is because the label space for these models is derived from the remapped DaCy annotations, which do not include these entity types in the dataset. As a result, the models were not exposed to these labels during fine-tuning and are therefore unable to predict them. This is a limitation of the current setup, which also emphasizes the need for manually annotated data aligned with our label schema. All things considered, the results indicate that

while XLM-RoBERTa models can capture entity types such as PER and LOC reasonably well, performance is constrained by limited data, label noise from remapping, and under-represented entity types.

5.5 Qualitative Analysis

To better understand the performance of the evaluated models, we conducted a qualitative error analysis on predictions from the different models. The analysis focuses on three main categories of errors: false positives, false negatives, and misclassifications.

5.5.1 Off-the-shelf NER model

False Positives False positives happen when the model predicts an entity where there is none in the gold data. One example is the token "Blomadisk", which was labeled as PER by both models (sm and lg), despite being annotated O in the gold data. This suggests that the models could incorrectly interpret uncommon or unknown words as person names. It appears there has been a speech-recognition error in this example, as this does not look like correct Danish, but we still decided to include this example for the sake of showing the performance.

Another example shows the token "VR",

Token	Gold Label	Predicted Label
Blomadisk	O	B-PER
nu	O	O
og	O	O
som	O	O
bliver	O	O
generende	O	O
hæbne	O	O
for	O	O
.	O	O

Table 21: Example of false positive: gold data vs. data labeled with the sm and lg models

which was classified as LOC by the smaller model (sm), while the large model (lg) correctly left it unlabeled. In this context, "VR" refers to "virtual reality" and not a location. This could mean that the smaller model overgeneralizes and assigns entity labels based on superficial patterns rather than contextual understanding.

Token	Gold Label	Predicted Label
Så	O	O
:	O	O
Øh	O	O
,	O	O
skal	O	O
du	O	O
have	O	O
en	O	O
introduktion	O	O
til	O	O
VR	O	B-LOC
versionen	O	O
af	O	O
det	O	O
?	O	O

Table 22: Example of false positive: gold data vs. data labeled with the sm model

Overall, false positives show a lack of domain awareness, where models misinterpret terms or uncommon words as named entities.

False Negatives False negatives happen when the model fails to detect an entity that is present in the gold data. One example is "rådhuset", which is labeled as ORG in the gold data, but missed by both the sm and lg

models. This indicates that the model could struggle with common nouns that can refer to institutions but are not always capitalized in the text.

Token	Gold Label	Predicted Label
Stadigvæk	O	O
i	O	O
i	O	O
rådhuset	B-ORG	O
.	O	O

Table 23: Example of false negative: gold data vs. data labeled with the sm and lg models

Another example is "organisk psykose", which is annotated as HEALTH but not detected by either model as MISC or anything else for the sake of anonymization. At least, this kind of error is expected as this fall outside the typical label space of the general purpose NER models such as spaCy, which is typically trained to detect entities like persons, organizations and locations.

Token	Gold Label	Predicted Label
Altså	O	O
jeg	O	O
har	O	O
jo	O	O
,	O	O
så	O	O
hedder	O	O
den	O	O
ikke	O	O
organisk	B-HEALTH	O
psykose	I-HEALTH	O
,	O	O
øh	O	O
.	O	O

Table 24: Example of false negative: gold data vs. data labeled with the sm and lg models

These errors show a fundamental limitation of the off-the-shelf NER models when applied to specialized domains, such as psychotherapy. Important entities relevant for anonymization can be systematically missed, as in our examples.

Misclassification Errors Misclassification happens when the model correctly identifies an entity span but it assigns it the wrong label.

In the example "Laksen", the gold annotation labels the token as MISC, while the smaller model classifies it as PER and the larger model does not assign any label. This suggests ambiguity in the token, which could be interpreted as either a surname or a noun, depending on the context.

Token	Gold Label	sm model	lg model
Fra	O	O	O
Laksen	B-MISC	B-PER	O
og	O	O	O
fra	O	O	O
Grønland	B-LOC	B-LOC	B-LOC
.	O	O	O

Table 25: Example of misclassification and missed entity: gold data vs. data labeled with the sm model vs. data labeled with the lg model

Another example is "Akrobat", which is labeled as MISC in the gold data, but classified as LOC by both models. This indicates that the models rely on learned patterns that may not generalize well to less common entity types.

Token	Gold Label	Predicted Label
På	O	O
Akrobat	B-MISC	B-LOC
.	O	O

Table 26: Example of misclassification: gold data vs. data labeled with the sm and lg models

Misclassification errors show limitations in both the models and challenges in the annotation scheme.

Across all errors types, there is a consistent pattern. The models perform reasonably well on standard named entities such as persons, locations and organizations but struggle with domain-specific and quasi-identifiers.

False positives often stem from overgeneralization, particularly in the smaller model, while false negatives are common for entities that fall outside the models' training distribution, such as health conditions and institutional

references. Misclassification errors just further highlight ambiguity in both the data and the label space.

5.5.2 Commercial PII detection system

False Positives False positives occur when the model predicts an entity that is not annotated in the gold data. The most common spurious labels are PER, DEM, and LOC.

For PER, the model tends to assign person labels to capitalised tokens even when they are not person entities. A simple example is *Orh*, a Danish interjection, which is incorrectly labelled as B-PER:

Token	Gold	Predicted
Orh	O	B-PER

Table 27: False positive: the interjection *Orh* is incorrectly tagged as PER.

Similar errors occur for tokens that appear to be speech recognition errors, such as *Bindelsen* and *klapsvøbner*. This follows the same pattern observed in the off-the-shelf NER models: unusual or capitalised word forms are sometimes enough to trigger a PER prediction. This suggests the model relies on surface form rather than conversational context.

For DEM, the model repeatedly labels isolated numbers from rating scales or approximate quantities (e.g. 17, 50, 99) as demographic entities, likely because they look like age expressions covered by the model's native *age* label.

False Negatives False negatives are much more common than false positives (68 cases for window-7), which indicates that low recall is the main weakness of the model. As shown in the quantitative results, many entity types receive an F1-score of zero because the label mapping does not capture how such entities appear in conversational psychotherapy data.

For HEALTH, clinical terms used in everyday speech are consistently missed. The word *angst* appears multiple times as a B-HEALTH

entity in the gold data, yet the model assigns 0 in each case:

Token	Gold	Predicted
en	O	O
angst	B-HEALTH	O
for	O	O

Table 28: False negative: *angst* (anxiety) is missed by the model.

For DEM, a recurring issue is that the model fails to detect multi-token spans. In the example below, the gold span *3 år* is annotated as a demographic age reference, but both tokens are predicted as 0:

Token	Gold	Predicted
siden	O	O
de	O	O
bliver	O	O
3	B-DEM	O
år	I-DEM	O
.	O	O

Table 29: False negative: the multi-token span *3 år* is entirely missed.

These examples suggest that the model struggles especially with domain-specific and conversational expressions, even when they are important for anonymization.

Misclassification Errors There are only 6 misclassification cases, and they remain the same across all window sizes, which suggests that adding more context does not help in these cases. The most common pattern is that MISC entities are predicted as PER. For example, the token *Laksen* is annotated as MISC in the gold data but predicted as PER, probably because it resembles a Danish surname:

The same pattern appears for *Tejer* and *Guggen*. Without a MISC label that adequately covers ambiguous cultural or contextual references, the model appears to fall back to the PER category.

A similar issue can be seen with *Loha*, a pseudonymous subject identifier annotated as

Token	Gold	Predicted
Fra	O	O
Laksen	B-MISC	B-PER
og	O	O
fra	O	O
Grønland	B-LOC	B-LOC
.	O	O

Table 30: Misclassification: *Laksen* (Gold: MISC) is predicted as PER. *Grønland* is correctly identified as LOC.

ID but predicted as PER. This is not surprising, since the model’s ID mapping mainly covers formal identifiers such as passport numbers or document IDs, rather than pseudonymous references used within transcripts.

Overall, the model performs reasonably on standard categories such as names and locations, but shows clear limitations when applied to psychotherapy transcripts. In particular, it struggles with domain-specific expressions, conversational language, and indirect identifiers, which are important to the anonymization task. As a result, its recall remains limited, reducing its practical usefulness in this domain.

5.5.3 SnakModel-7B-Instruct

False Positives After taking a closer look at false positive predictions made by the SnakModel, it became clear that the MISC category was the cause of the vast majority of incorrect detections, scoring 672 cases out of a total of 908 false positives. Labels PER and LOC contributed an additional 118 and 76 false positives, respectively, while the remaining entity types appeared far less frequently. This suggests the model frequently mislabeled ordinary words that carried no sensitive information in the gold standard. Many of the false positives originated from short expressions or common nouns being incorrectly interpreted as entities. This behaviour was especially visible for the MISC label, where the model frequently overgeneralized.

One such example of a false positive predic-

Token	Gold	Predicted
En	O	O
bog	O	B-MISC
Yes	O	O
!	O	O

Table 31: Example of a false positive prediction where the model incorrectly classified a non-entity as MISC.

tion in a sentence “*En bog Yes!*”, as seen in Table 31, where the model incorrectly tagged “*bog*” as a MISC entity, but despite this, no entity was present in the gold annotation. The model’s tendency to over-predict contextually insignificant words is a pattern that occurred frequently throughout the evaluation and contributed to the high false positive count.

Token	Gold	Predicted
Som	O	O
Jo	O	B-PER
er	O	O
et	O	O
forskningsprojekt	O	B-MISC
.	O	O

Table 32: Example of a false positive prediction where the model incorrectly classified a non-entity as PER.

In another example, seen in Table 32, the model mistakenly tagged “*Jo*” as a PER entity, but this is incorrect as there was no entity at this token in the gold label data. In this context “*Jo*” functions as a common conversational word used for emphasis or confirmation rather than a personal name. An example like this indicates that the model occasionally relied on surface-level cues such as capitalization rather than contextual meaning when predicting entities and, therefore, made incorrect predictions.

False Negatives The false negative analysis revealed that the model failed to detect a total of 44 entities present in the gold annotations. Missed detections were most common in the DATE and DEM label categories, with a total of 10 such cases for both, and followed closely by HEALTH with 9 cases. The remaining missed entities were spread across MISC,

LOC, ID, and ORG.

Many of the missed entities fell within the DATE and ID categories, where annotations frequently consisted of brief numerical or time-related expressions embedded into everyday conversation. Both HEALTH and DEM entities tended to go undetected when phrased indirectly or embedded within informal spoken language.

Token	Gold	Predicted
Dag	O	O
har	O	O
vi	O	O
halvanden	B-DATE	O
time	I-DATE	O
.	O	O

Table 33: Example of a false negative prediction where the model failed to detect a DATE entity present in the gold annotations.

As seen in Table 33, the expression “*halvanden time*” was annotated as a DATE entity in the gold data, yet the model did not label any entity whatsoever. The occasional difficulty in picking up on temporal expressions by the model can be clarified by the form of the entity, which appeared as part of natural speech rather than in a clearly structured format.

The phrase “*big community gruppe agtig*”, shown in Table 34, displays an interesting case of a missed entity. The sentence is blending English into Danish and is delivered in an informal manner of spoken language. In the gold annotation it was marked as a MISC entity, however the model did not label this. The combination of language-switching and conversational looseness likely made it difficult for the model to recognize the phrase as carrying any sensitive or entity-relevant meaning, and so it was overlooked.

Misclassification Errors There were 24 identified cases, in which the model managed to locate a relevant entity span but failed when it came to assigning the correct label. The most frequent confusion appeared when DEM was

Token	Gold	Predicted
Jeg	O	O
:	O	O
Jeg	O	O
blev	O	O
lidt	O	O
glad	O	O
for	O	O
jeg	O	O
skulle	O	O
i	O	O
den	O	O
der	O	O
big	B-MISC	O
community	I-MISC	O
gruppe	I-MISC	O
agtig	I-MISC	O
.	O	O

Table 34: Example of a false negative prediction where the model failed to identify a multi-token MISC entity present in the gold annotations.

misclassified as MISC, accounting for 4 cases. This was followed by DEM being classified as PER and ID being classified as PER, with 3 and 2 cases, respectively. Additionally, several entities were incorrectly assigned to the broader MISC category.

The main issue seems to be the model’s difficulty in distinguishing between semantically close entity types. Demographic terms that appear in conversational speech, numerical terms, and phrases were most commonly mixed up with broader categories such as MISC or PER, which suggests that the boundaries between these labels were not always clear enough for the model to resolve reliably.

One example of such misclassification can be seen in Table 35, where the expression “5 søskende” was annotated by the model as MISC but was correctly supposed to be labeled as a DEM entity according to the gold data. The model recognized the span of the entity as sensitive content but did not manage to capture the demographic character in this context and instead resorted to a more general classification. This case can be seen as an example of

Token	Gold	Predicted
Tænker	O	O
:	O	O
Måske	O	O
hvis	O	O
jeg	O	O
havde	O	O
5	B-DEM	B-MISC
søskende	I-DEM	I-MISC
.	O	O

Table 35: Example of a misclassification where a demographic entity was incorrectly predicted as MISC instead of DEM.

semantic overlap between the labels, which led the model to misclassifying this instance.

5.5.4 Fine-tuned transformed-based model

False Positives False positive errors made by the XLM-RoBERTa models were related to how numerical values were treated during the preprocessing step. In Table 36, it is shown how the expression “100%” is incorrectly labeled as MISC by all the models, even though it is annotated as O in the gold data. This model behavior can be attributed to the label mapping applied to the transcribed data. The automatically transcribed data came with labels such as CARDINAL, ORDINAL, and PERCENT, which were relatively frequent. To align these with our label set, they were mapped to MISC, as they did not fit with any of the other label categories (4). Since the label set does not include a dedicated category for general numeric values, the models learn to associate numbers with the MISC label during fine-tuning.

However, this leads to a mismatch with the gold dataset, where numeric values were not annotated as entities unless they correspond to sensitive information such as IDs, CPRs, or contact details. As a consequence, the XLM-RoBERTa models systematically over-predict MISC for any numeric token, which contributes to lower performance for this class

in the quantitative evaluation (refer to big table). The potential concern is that more sensitive numeric expressions (e.g. some type of ID) may be classified as MISC rather than their intended class, which would limit the effectiveness of the pseudonymization step.

Token	Gold	Base	Large	Large + LoRA
100	O	B-MISC	B-MISC	B-MISC
%	O	I-MISC	I-MISC	O
overbevist	O	O	O	O
og	O	O	O	O
jeg	O	O	O	O
er	O	O	O	O
et	O	O	O	O
godt	O	O	O	O
menneske	O	O	O	O
.	O	O	O	O

Table 36: Example of false positive: the numeric expression “100 %” is incorrectly labeled as MISC by all models, with Base and Large predicting a multi-token span while LoRA predicts only a partial span.

Token	Gold	Base	Large	Large + LoRA
Vi	O	O	O	O
kan	O	O	O	O
genbesøge	O	O	O	O
og	O	O	O	O
skabe	O	O	O	O
den	O	O	O	O
samme	O	O	O	O
situation	O	O	O	O
2	O	B-MISC	B-MISC	B-MISC
gange	O	O	O	O
.	O	O	O	O

Table 37: Example of false positive: the numeric token “2” is incorrectly labeled as MISC by all models, despite not representing a named entity in context.

In a similar manner, Table 37 shows how the token “2” in the phrase is incorrectly labeled as MISC by all models. Unlike the first example, this token appears in a linguistic context, where it functions as a quantity (“2 gange”), yet the models still assign an entity label due to its numeric form, rather than its meaning in the sentence. As previously mentioned, this suggests that the models have developed a bias toward labeling any numeric value as MISC, likely to their frequency in the fine-tuning data. From the perspective of pseudonymization,

general numerical expressions are not considered sensitive and would not require masking, as no identifiable information is disclosed. Even in cases where such phrase refers to a medical context, such as dosage instructions (“2 gange om dagen”) or medication quantity (“50 mg Setralin”), it would still not constitute sensitive information needing anonymization.

For the XLM-RoBERTa models, false positives were largely due to systematic biases introduced during the preprocessing and label mapping, particularly the use of MISC as a broad class. While this design choice simplifies the label set, it also affects the model’s ability to distinguish between sensitive IDs and general numerical expressions. In principle, numerical values could have been excluded from the MISC class, however due to the limited amount of annotated data, this mapping was retained to see its effect on the model behavior. To address these limitations, a more rule-based method for defined numerical values could be implemented e.g. through regular expressions.

False Negatives In Table 38 it is shown that none of the models identify the token “Opus” as an entity, despite it being annotated as HOSPITAL. In this particular context, “Opus” refers to a specialized treatment program for patients with psychosis offered by the Danish psychiatric system. Such information can be considered sensitive since there are only about 2.300 patients annually getting treated through OPUS (Region Hovedstadens Psykiatri, 2025). The failure to identify this type of entities may increase the risk of re-identification of patients. This error is likely due to the low representation of domain-specific institution names as HOSPITAL or ORG in the fine-tuning data. As a consequence, the models default to predicting non-entity labels O, even when the context of the sentence indicates a sensitive reference such as this hospital.

Similarly in Table 39, the token “koraner” is

Token	Gold	Base	Large	Large + LoRA
Altså	O	O	O	O
sådan	O	O	O	O
henne	O	O	O	O
i	O	O	O	O
Opus	B-HOSPITAL	O	O	O
for	O	O	O	O
at	O	O	O	O
få	O	O	O	O
behandling	O	O	O	O
det	O	O	O	O
sådan	O	O	O	O
.	O	O	O	O

Table 38: Example of false negative: the institution name "Opus" (Gold: HOSPITAL) is not detected by any of the models.

annotated as RELIGION in the Gold dataset but is not identified by any model. This entity does not refer to a religion but to the religious text of Islam, the Quran. While such reference itself is not directly revealing, it can contribute to identity disclosure when combined with several quasi-identifiers. For instance, if it is known that the patient is undergoing treatment through OPUS and is associated with a minority religious background in Denmark, this information together could disclose the identity of the patient due to the relatively small number of patients. Ideally, the expectation would be that models identify this as the parent category DEM or even MISC since they are present in the fine-tuning data. However, the absence of a label given by the models suggests there is a need for more samples for DEM-related entities. In general, domain-specific entities and less-frequent labels are more likely to be missed posing a challenge for pseudonymization.

Misclassification Errors The misclassification errors observed for the fine-tuned models reflect some semantic overlap between entity categories and limitations present in the training data.

As shown in Table 40, the span "20 minutter frem" is annotated as duration in the gold data, yet all models have difficulties capturing both its label and full span. The base and large

Token	Gold	Base	Large	Large + LoRA
Så	O	O	O	O
jeg	O	O	O	O
sådan	O	O	O	O
,	O	O	O	O
Det	O	O	O	O
var	O	O	O	O
nogle	O	O	O	O
smukke	O	O	O	O
koraner	B-RELIGION	O	O	O
der	O	O	O	O
.	O	O	O	O

Table 39: Example of false negative: the token "koraner" (Gold: RELIGION) is not identified as an entity by any of the models.

models classify the first two tokens as TIME, while the LoRA-adapted model predicts DATE. All models failed to include the token "frem" as part of the entity. This can suggest that multi-token entities might be underrepresented in the training data.

Although these predictions are not surprising, given that duration, time, and date are all temporal expressions usually involving numbers and strings, they still show that models cannot distinguish between semantically-close entities. This problem is further influenced by the design of the label set itself. If a single and general label such as TEMPORAL were used to group these tags together, these inconsistencies would disappear. However, such a simplification would come at the cost of less granularity between time-related expressions. In a pseudonymization context, this granularity is important, as these different tags require different types of surrogate replacement in the anonymization pipeline. In other words, a single TEMPORAL tag would make it difficult to create contextually appropriate substitutions.

Another example is observed in Table 41, where the span is correctly identified, but the assigned label is incorrect. The token "Juleaften" is annotated as a DEM entity in the gold data, yet both the base and large models classify it as MISC, while the LoRA model labels it to DATE. The tendency to assign the more broad MISC label may reflect the im-

Token	Gold	Base	Large	Large + LoRA
Jeg	O	O	O	O
har	O	O	O	O
ingen	O	O	O	O
brug	O	O	O	O
for	O	O	O	O
20	B-DURATION	B-TIME	B-TIME	B-DATE
minutter	I-DURATION	I-TIME	I-TIME	I-DATE
frem	I-DURATION	O	O	O
og	O	O	O	O
⋮	⋮	⋮	⋮	⋮

Table 40: Example of misclassification: the duration expression “20 minutter frem” is incorrectly labeled as TIME (Base, Large) and DATE (LoRA), with incomplete span detection.

Token	Gold	Base	Large	Large + LoRA
Juleaften	B-DEM	B-MISC	B-MISC	B-DATE
eller	O	O	O	O
en	O	O	O	O
fødselsdag	O	O	O	O
,	O	O	O	O
hvor	O	O	O	O
jeg	O	O	O	O
ikke	O	O	O	O
er	O	O	O	O
...	...	...	...	...

Table 41: Example of misclassification: the entity “Juleaften” (Gold: DEM) is consistently mislabeled as MISC (Base, Large) and as DATE (LoRA), despite correct span detection.

balance in the fine-tuning data, where less-common labels like DEM are harder for the models to learn.

The token "Juleaften" corresponds to the mention of the holiday period (Christmas Eve), which can be ambiguous, since it might be interpreted both as a cultural event (DEM) and as a calendar-based reference (DATE). While this instance does not reveal any sensitive information, since Christmas is a national holiday in Denmark, the same type of misclassifications could have worse implications in another context. For example, if there was a reference to a less common and minority religious event, it could contribute to indirect identification in combination with other quasi-identifiers.

All together, these examples illustrate that misclassifications are not always random, but can arise due to systematic challenges, such as semantically-overlapping tags and low rep-

resentation of certain entity types. While the models capture general patterns to some degree, they struggle with correctly identifying fine-grained labels.

5.6 Anonymization Output Examples

To illustrate the behavior of the anonymization pipeline, this section presents selected examples of pseudonymized outputs. The examples show how the pipeline transforms different entity types depending on the pseudonymization strategy assigned to each tag (see Table 13).

Original	Pseudonymized	Tag
<i>Jeg hedder Julie.</i>	<i>Jeg hedder Mette.</i>	PER
<i>I Tønder også.</i>	<i>I [dansk by] også.</i>	LOC
<i>Og hun har 5 gange.</i>	<i>Og hun har [misc info] gange.</i>	MISC
<i>Vi kan genbesøge situationen 10 gange.</i>	<i>Vi kan genbesøge situationen [misc info] gange.</i>	MISC
<i>Sådan slovakisk skal han.</i>	<i>Sådan [demografisk info] skal han.</i>	DEM

Table 42: Illustrative examples of pseudonymized transcript output. *Note:* Examples are illustrative and do not originate from real patient data.

The examples illustrate several properties of the pipeline. Person names are replaced with surrogate Danish first names drawn from a fixed pool, meaning the same surrogate may appear for different individuals within a session. Location references such as city names are replaced with the fixed placeholder [dansk by], regardless of the specific city mentioned. Numeric expressions flagged as MISC, such as counts or quantities, receive the generic placeholder [misc info], which preserves the grammatical structure of the sentence while removing the potentially identifying value. Demographic attributes such as language or nationality references are replaced with [demografisk info].

It is worth noting that the pipeline also produces errors that are visible in the output. In several cases, DATE predictions contain false positives where ordinal expressions such as *tredje* or weekday references such as *fredags*

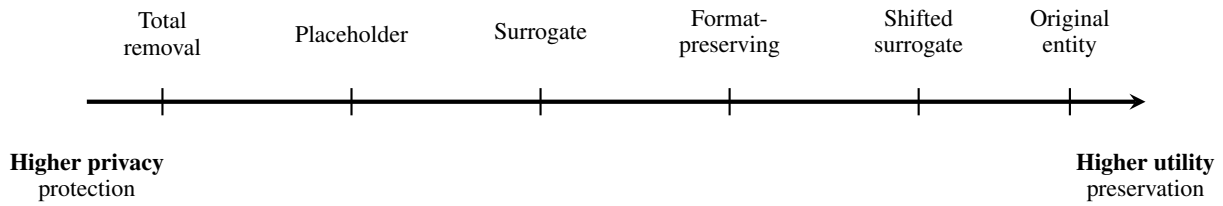


Figure 7: Illustration of the privacy-utility trade-off for different pseudonymization strategies (Table 44) used in the pipeline. Strategies further to the left prioritize stronger privacy protection, while strategies further to the right preserve more semantic, structural, or contextual utility in the text.

are incorrectly identified as dates and replaced with shifted date values (e.g. *19.10.2009*), resulting in unnatural output. This highlights that the quality of the pseudonymized text is directly dependent on the accuracy of the underlying NER predictions, and that false positives in entity detection translate into visible artifacts in the anonymized transcript.

5.7 Reflections on Pseudonymization Strategies

The pseudonymization strategies used in this project reflect a broader trade-off between privacy protection and data utility (Yermilov et al., 2023). Different replacement methods preserve different amounts of semantic and contextual information, and there is no one optimal solution for all cases. At one extreme, leaving an entity unchanged preserves the full utility of the original text, providing no privacy protection. At the other extreme, fully removing entities maximizes privacy, but severely reduces analytical usefulness. The strategies used in the project attempt to balance these two ends by selectively preserving certain types of information while obscuring identifying details. Figure 7 illustrates how the different pseudonymization strategies used in the pipeline can be positioned along this privacy-utility axis, while the complete mapping of entity tags to replacement methods is shown in Table 44.

For instance, placeholders such as [dansk by] preserve rough semantic information about the entity category while hiding the orig-

inal value. In this way, the surrounding sentence remains interpretable without revealing the specific location. Similarly, surrogate replacements with random names, relations, or conditions preserve more of the readability and flow of the text by replacing entities with plausible alternatives rather than just a generic placeholder. However, this also introduces challenges related to consistency and realism. Format-preserving replacements occupy an intermediate position in the privacy-utility trade-off shown in Figure 7. These methods preserve structural patterns such as the formatting of phone numbers, postcodes, email addresses, or identifiers, which are relevant to Denmark, while still replacing the original sensitive values. Shifted surrogates are used for dates and times to preserve partial temporal relationships while hiding the exact values. Compared to surrogates, this retains more contextual information about the sequence and timing of events, but may also increase the risk of preserving potentially identifying temporal patterns.

A major drawback of the current pipeline is that it does not track entities across the transcript. As a result, the same individual may be replaced with different surrogate names across the same transcript, while multiple distinct individuals may accidentally receive the same replacement. This reduces the coherence of the text. To address this problem, it would require methods related to co-reference resolution or similar techniques, where the pipeline would maintain consistent mappings between mentions throughout a document (Zheng et al.,

2011; Lee et al., 2017). Such approaches were outside the scope of this project.

Another drawback concerns the evaluation of the pseudonymization pipeline. While this project primarily evaluates entity detection performance using standard NER metrics such as precision, recall, and F1-score, these metrics do not directly translate to privacy risk. In reality, the effectiveness of pseudonymization depends not only on whether entities are detected, but also on whether individuals remain re-identifiable through combinations of remaining information and external background knowledge (Ohm, 2010). Evaluating such risks is closer to a privacy and security problem rather than a traditional NLP task. Therefore, in this project, the effectiveness of pseudonymization remains dependent on the quality of the underlying NER task. This raises the question of whether NER alone is sufficient for anonymizing psychotherapy transcripts. Originally, NER systems were developed for extracting structured entities from domains such as journalistic articles (Grishman and Sundheim, 1996; Munnangi, 2024). Psychotherapy transcriptions are fundamentally different in their conversational structure, e.g., incomplete sentences, implicit references, and context-dependent personal information. In other words, potentially identifying details in psychotherapy data are not explicitly named entities but emerge implicitly through a combination of events, relationships, life circumstances, etc. As a consequence, framing pseudonymization purely as an NER problem may overlook important aspects of solving this task.

All in all, these reflections suggest that psychotherapy transcripts may require methods that extend beyond standard sequence labeling. Rather than viewing pseudonymization as a purely NER problem, it may be more appropriate to treat it as a broader privacy risk assessment task involving legal, contextual, and

domain-specific considerations.

6 Discussion

Evaluation metrics We found that the evaluation of anonymization systems in domain-specific settings such as psychotherapy is more complex than standard performance metrics suggest. While our models achieve measurable precision, recall, and F1-scores, these metrics alone do not fully capture the effectiveness of our anonymization. In particular, the cost of errors. False negatives, where sensitive entities are not detected, directly translate to potential privacy leakage. False positives, where entities are mistakenly labeled, primarily affect data utility. This shows that standard evaluation metrics can be misleading, because they hide important risks that actually matter in practice for this type of problem.

Comparisons Furthermore, our comparison of different models also has some important limitations. The systems we evaluate differ in architecture, training data, label space, and degree of domain adaptation. For example, off-the-shelf models trained on Danish news text operate under very different assumptions than fine-tuned or few-shot prompt approaches designed for this task. In addition, having to map our detailed annotation labels into a smaller set of categories introduced information loss and simplified the problem. As a consequence, the differences observed in performance should be interpreted with caution, as they may reflect differences in setup rather than true differences in model capability.

Data size Another important point is the size and composition of the evaluation data. The gold dataset is based on a single manually annotated psychotherapy session, which limits how reliable our results are. With such a small sample, even a few annotation or prediction errors can noticeably affect the evaluation metrics. A further complication would be class

imbalance, as most tokens are labeled as non-entities. As a result, even very simple strategies, such as predicting everything as „O“ can achieve high scores on some metrics while being practically useless. This shows the need to interpret our results in context and to consider appropriate baselines and error analysis.

Biases Beyond these technical limitations mentioned above, our work is also influenced by cognitive biases in the research process. The way we designed the annotation schema, annotated the data, and evaluated the models is closely connected, which introduces a risk of confirmation bias. In other words, we may be evaluating the models based on assumptions we defined ourselves. In addition, relying on existing benchmarks to design the tagset may have shaped our understanding of what counts as a relevant entity, potentially causing us to miss more subtle, domain-specific identifiers that can be present in psychotherapy data. This shows that the model performance is not only determined by the technical background, but also by how the task is defined (Ho and Beyan, 2020).

Furthermore, our results show the impact of algorithmic bias caused by differences in the training data. Models trained on Danish news text or legal domains struggle with the language context of psychotherapy transcripts, which leads to predictable errors such as over-generalizing or miss context-dependent entities. This shows that the limitations we observe are not random, but come from the type of data the models were trained on. It also emphasizes the importance of adapting models to the specific domain (Ho and Beyan, 2020).

Ethical concerns Finally, our findings raise important fairness and ethical concerns. The model does not protect all types of information equally, as some entities, such as health-related or context-dependent identifiers, are detected less reliably. This means that individuals may

not be equally protected by the anonymization process, depending on how their identity appears in the text. Given the sensitive nature of psychotherapy data, this raises the question of whether this anonymization is sufficient to be used.

Overall, these problems show that while NER-based anonymization is a promising approach, its application is sensitive, domain-specific, and requires careful consideration of evaluation practices, data constraints, and ethical implications.

7 Limitations

7.1 Data Constraints

An important limitation is the quality and quantity of the available data. The majority of the data consists of automatically transcribed psychotherapy sessions, which introduce errors in the form of transcription errors, repeated or nonsensical phrases, and missing or incorrectly recognized words due to the limitations of the speech-to-text model. Such an example would be the transcription model misspelling names (e.g. Berta -> Bertha), which would be a complication for the pseudonymization process, where we would ideally want to keep track of the person and replace them with the same surrogate value across the text. Additionally, the nature of the data leads to frequent short utterances, fillers, pauses, and incomplete sentences, which further complicates the annotation process. At the time of writing this, only a small subset of the data is available for supervised use, which is 12 sessions annotated automatically and one session manually annotated by us. The automatically annotated sessions may have incorrect or incomplete labels and do not fully align with our defined tagset (as defined in Table 4). The manually annotated data is limited in size (around 100 tags) and may still include human error. These constraints result in a dataset that is both noisy and has few labels, limiting the ability to effectively

train or fine-tune models and likely contributing to their low performance, especially for less frequent entities. Furthermore, there is a significant class imbalance, where certain tags appear often while others only rarely, making it difficult for models to learn robust patterns across all labels. This imbalance is reflected in uneven performance across entity types and reduces the reliability of the metrics. In addition, annotation inconsistency shows up from the combination of automatic and manual labeling processes, which follow different criteria and lead to conflicting signals. Finally, the definition of some indirect identifiers is inherently ambiguous, as these entities often depend on context and subjective interpretation. This ambiguity makes consistent annotation challenging for human annotators and introduces uncertainty into both model training and evaluation.

7.2 Legal Constraints

Another limitation of this project is the legal constraint imposed by GDPR, which restricts the handling of sensitive data such as psychotherapy transcripts. Strict requirements must be followed to ensure that individuals in the dataset cannot be identified, either directly or indirectly. This was particularly problematic in designing the project when deciding on the tagset (Table 4). GDPR emphasizes the need to account for not only direct identifiers but also indirect identifiers, which increases the complexity of the task. With no legal expertise available to review our tagset, we cannot guarantee that our proposed schema fully captures all relevant identifiers required for proper anonymization. This introduces uncertainty in both the completeness and correctness of our approach, as certain quasi-identifiers may have been overlooked or insufficiently defined. Additionally, the interpretation of what is an indirect identifier can vary depending on context, making it difficult to establish clear and

universally applicable annotation guidelines. These constraints also limit the ability to validate the anonymization process against legal standards. As a result, this project operates under a degree of legal uncertainty, where ensuring compliance and completeness remains an open challenge.

7.3 Technical Constraints

Other limitations of this project are technical and infrastructure constraints related to working with sensitive data in a secure environment. Due to data protection requirements, all data processing and experimentation had to be done on the hospital-provided laptops, which limited both computational capacity and flexibility. Access to laptops was also initially restricted, with only one available during the first weeks of the project, gradually increasing over time. Furthermore, until mid-March, no cloud computing resources were available, making it difficult to run larger models or conduct extensive experiments. During this period, access to the data was limited to on-site work at the hospital, typically once a week, significantly slowing down development since we could not bring the hospital-provided laptops home. In addition, the data itself was provided incrementally, with early versions consisting of low-quality transcriptions, followed by improved versions from a different model, and delays in receiving manually transcribed and annotated data. This fragmented and evolving setup introduced inconsistencies and hindered systematic experimentation, impacting the efficiency of our model development and evaluation.

8 Conclusion and Future Work

In this project, we investigated whether a pipeline combining NER and rule-based pseudonymization can effectively anonymize Danish psychotherapy transcripts in order to use them for further LLM training. To do this,

we compared four approaches representing different levels of domain adaptation: off-the-shelf NER model (spaCy), a commercial PII detection system (EU PII Safeguard), a few-shot prompting approach (SnakModel), and a fine-tuned model (XLM-RoBERTa).

Unfortunately, our results showed that no single approach fully solves this problem. The fine-tuned XLM-RoBERTa achieved the best overall performance, with the large variant performing best on person entities. The commercial EU PII Safeguard model was surprisingly a good competition, while the off-the-shelf spaCy models and the few-shot prompting approach struggled to capture the domain-specific entities. For all approaches, performance was good on standard categories such as PER and LOC, but consistently weak on domain-specific quasi-identifiers such as HEALTH, HOSPITAL, DEM, etc.

To answer our research question, we can say that while the NER and rule-based pseudonymization pipeline is a viable framework for anonymizing psychotherapy transcripts, the current models do not yet reach a level of reliability sufficient for unsupervised use on sensitive clinical data. The pipeline successfully transforms detected entities into pseudonymized text that preserves linguistic structure for LLM training, but its effectiveness is fundamentally limited by the underlying detection step. As long as domain-specific identifiers are missed, the pipeline cannot guarantee the level of privacy protection required by GDPR.

There are several directions we could go for future work. First and foremost, expanding the manually annotated dataset is essential, since our evaluation relied on a single annotated session. The fine-tuning corpus was derived from automatic DaCy annotations remapped to our schema, which introduced noise and limited the model’s ability to learn underrepresented entity types. A large, manually annotated cor-

pus aligned directly with our tag scheme would improve both training and evaluation reliability. Secondly, we could try combining models rather than relying on a single one. An ensemble approach that combines the strong recall of EU PII Safeguard for PER with the domain adaptation of the fine-tuned XLM-RoBERTa models could improve coverage across entity types. Thirdly, we could reevaluate our evaluation practices. We could move beyond standard precision, recall, and F1 scores towards metrics that explicitly weigh the cost of false negatives, since missed entities carry far greater real-world risk than false positives. Finally, the legal validity of the anonymization pipeline should be assessed in collaboration with legal experts to ensure that the proposed tag scheme and pseudonymization strategies meet GDPR requirements in practice.

All in all, this work contributes a decent evaluation of anonymization approaches in a highly sensitive domain, along with a reusable pipeline that supports the broader goals of the ALF project. While further work is required before such a system can be deployed in reality, our findings provide a foundation for developing more reliable anonymization methods for mainly Danish, but also any language clinical text.

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A EU PII Safeguard Label Mapping

EU PII Safeguard Label	Mapped Label
B-ACCOUNT_NUMBER	B-ID
B-ADDRESS	B-LOC
B-AGE	B-DEM
B-AMOUNT	B-MISC
B-BUILDING_NUMBER	B-LOC
B-CITY	B-LOC
B-COMPANY_NAME	B-ORG
B-COUNTRY	B-LOC
B-CREDIT_CARD_NUMBER	B-ID
B-CURRENCY	B-MISC
B-DEVICE_ID	B-MISC
B-DOB	B-DATE
B-DRIVER_LICENSE	B-ID
B-EMAIL	B-CONTACT
B-ETHNICITY	B-DEM
B-FIRSTNAME	B-PER
B-GENDER	B-DEM
B-HEALTH_CONDITION	B-HEALTH
B-HEALTH_INSURANCE_ID	B-ID
B-IBAN	B-ID
B-IP_ADDRESS	B-MISC
B-JOB_TITLE	B-DEM
B-LASTNAME	B-PER
B-LATITUDE	B-LOC
B-LONGITUDE	B-LOC
B-MAC_ADDRESS	B-MISC
B-MIDDLENAME	B-PER
B-NATIONAL_ID	B-ID
B-PASSPORT_NUMBER	B-ID
B-PASSWORD	B-MISC
B-PHONE_NUMBER	B-CONTACT

EU PII Safeguard Label	Mapped Label
B-POLITICAL_OPINION	B-DEM
B-POSTAL_CODE	B-LOC
B-PREFIX	B-PER
B-RELIGION	B-DEM
B-SALARY	B-MISC
B-SEXUAL_ORIENTATION	B-DEM
B-STATE	B-LOC
B-STREET	B-LOC
B-TAX_ID	B-ID
B-URL	B-MISC
B-USERNAME	B-PER
I-ADDRESS	I-LOC
I-AMOUNT	I-MISC
I-BUILDING_NUMBER	I-LOC
I-CITY	I-LOC
I-COMPANY_NAME	I-ORG
I-COUNTRY	I-LOC
I-DOB	I-DATE
I-EMAIL	I-CONTACT
I-FIRSTNAME	I-PER
I-GENDER	I-DEM
I-HEALTH_CONDITION	I-HEALTH
I-HEALTH_INSURANCE_ID	I-ID
I-IP_ADDRESS	I-MISC
I-JOB_TITLE	I-DEM
I-LASTNAME	I-PER
I-LATITUDE	I-LOC
I-LONGITUDE	I-LOC
I-MAC_ADDRESS	I-MISC
I-MIDDLENAME	I-PER
I-NATIONAL_ID	I-ID
I-PASSPORT_NUMBER	I-ID
I-PASSWORD	I-MISC

EU PII Safeguard Label	Mapped Label
I-PHONE_NUMBER	I-CONTACT
I-POSTAL_CODE	I-LOC
I-PREFIX	I-PER
I-SALARY	I-MISC
I-STATE	I-LOC
I-STREET	I-LOC
I-TAX_ID	I-ID
I-URL	I-MISC
I-USERNAME	I-PER
O	O

Table 43: Complete mapping from EU PII Safeguard labels to the shared evaluation label space. BIO prefixes (B-, I-) are preserved during mapping.

B Pseudonymization Strategy Mapping

Tag	Replacement type	Description
PER	Surrogate	Random name drawn from a pool of 10 common Danish first names
STAFF	Surrogate	Role-bearing form Psykolog <Name>, where the name is drawn from the same pool of Danish first names
PATIENT	Surrogate	Random name drawn from the same pool as PER
RELATION	Surrogate	Random family relation drawn from a pool of 9 Danish kinship terms (e.g. <i>mor</i> , <i>far</i> , <i>søster</i>)
LOC	Placeholder	Fixed placeholder [dansk by] (city in Danish)
ADDRESS	Placeholder	Fixed placeholder [dansk adresse] (address in Danish)
POSTCODE	Format-preserving	Random Danish postcode in XX00 format
ORG	Placeholder	Fixed placeholder [organisation] (organization in English)
HOSPITAL	Placeholder	Fixed placeholder [dansk hospital] (Danish hospital in English)
PHONE	Format-preserving	Random number in +45 XXXXXXXX format
EMAIL	Format-preserving	Random email address in name@gmail.com format, where the name is drawn from the name pool
CONTACT	Format-preserving	Randomly replaced with either a PHONE or EMAIL surrogate
ID	Format-preserving	Random 10-digit identifier
CPR	Format-preserving	Random 10-digit CPR-like identifier
DATE	Shifted surrogate	Original date shifted by a random offset (± 1 –30 days and 1–12 months); falls back to a random date in 2000–2026 if unparseable
TIME	Shifted surrogate	Original time shifted by ± 1 –24 hours and 1–59 minutes; falls back to a random time if unparseable
DURATION	Surrogate	Random number substituted into the original expression, preserving the time unit (e.g. <i>10 gange</i> $\rightarrow$ <i>7 gange</i> , <i>3 år</i> $\rightarrow$ <i>5 år</i>)
DEM	Placeholder	Fixed placeholder [demografisk info] (demographic information in English)
ETHNICITY	Placeholder	Fixed placeholder [etnicitet] (ethnicity in English)

Tag	Replacement type	Description
RELIGION	Placeholder	Fixed placeholder [religion] (religion in English)
POLITICS	Placeholder	Fixed placeholder [politik] (politics in English)
SEXUALITY	Placeholder	Fixed placeholder [seksualitet] (sexuality in English)
AGE	Surrogate	Random age drawn uniformly from an 8-year age interval within the range 18–89
HEALTH	Placeholder	Fixed placeholder [helbredsoplysninger] (health information in English)
DIAGNOSIS	Placeholder	Fixed group label [schizofrenispektrumdiagnose] (schizophrenia spectrum diagnosis in English)
MEDICATION	Placeholder	Fixed group label [antipsykotisk medikation] (antipsychotic medication in English)
CONDITION	Surrogate	Random condition drawn from a pool of 5 (e.g. depression, diabetes)
MISC	Placeholder	Fixed placeholder [diverse info] (miscellaneous information in English)

Table 44: Complete pseudonymization strategy for each entity tag. Pool sizes and examples reflect the implementation used in this project.